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Systems and Methods For Assembling Liquid Desiccant Air Conditioner Panels Using Flexible Alignment Features

Abstract

A method of assembling a multilayer panel for use in a heat exchanger includes positioning a frame on a work platform, the frame having two header sections and a middle section, the middle section defining a heat transfer fluid area and each header section defining a header area. The method also includes positioning a heat exchange sheet on the frame across the middle section and each header section, welding the heat exchange sheet to the frame to form a middle weld enveloping the heat transfer fluid area and two outer welds each enveloping a corresponding one of the header areas, and controlling alignment between the frame and the heat exchange sheet during welding.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a divisional of U.S. patent application Ser. No. 18/390,475, filed on Dec. 20, 2023, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD

[0002] The field relates generally to heating, ventilation, and air conditioning (HVAC) systems, and more particularly, to systems and methods for assembling multilayer panels (or panel assemblies) used in a three-way heat exchanger operable to transfer heat between a heat transfer fluid, a liquid desiccant, and air.

BACKGROUND

[0003] Heating, ventilation, and air conditioning (HVAC) systems are known for their heating, cooling, and moisture removal capabilities for treating outside air that is circulated through an indoor space. The vapor compression cycle is widely used in HVAC systems to regulate the temperature and humidity of the outside air. Typically, outside air is cooled below its dew point temperature to allow moisture in the air to condense on an evaporator coil, thus dehumidifying the air. Since this process often leaves the dehumidified air at an uncomfortably cold temperature, the air is then reheated to a temperature more comfortable to a user. The process of overcooling and reheating the air can become very energy-intensive and costly.

[0004] In some applications, HVAC systems include a vapor compression system used in combination with a liquid desiccant dehumidification system to remove moisture from the outside air without cooling it below its dew point temperature. For example, HVAC systems may include a refrigerant sub-system that operates under the vapor compression cycle and an air treatment sub-system that uses heat transfer fluid and liquid desiccant to simultaneously absorb heat (sensible cooling) and moisture (latent cooling) from warm outside air to produce cooled and dehumidified indoor air. The air treatment sub-system may include three-way heat transfer equipment that facilitates sensible and latent cooling of the warm outdoor air using the heat transfer fluid and the liquid desiccant.

[0005] In operation of a three-way heat exchanger, the liquid desiccant and heat transfer fluid are channeled through the heat exchanger and heat is transferred between the liquid desiccant and the heat transfer fluid. An outdoor air stream is directed through the heat exchanger, and heat transfer fluid absorbs heat from the air stream while the liquid desiccant absorbs moisture from the air stream. The liquid desiccant may circulate between the three-way heat exchanger and a regeneration system, in which diluted liquid desiccant rejects the absorbed moisture into a sacrificial fluid. The refrigerant sub-system interfaces with the air treatment sub-system, whereby refrigerant in an evaporation stage of the vapor compression cycle absorbs heat from the heat transfer fluid in the three-way heat exchanger. The refrigerant is then channeled to a condensing stage in which the refrigerant rejects the absorbed heat into another fluid. Liquid desiccant treated by the regeneration system and heat transfer fluid treated by the refrigerant sub-system is then channeled back toward the three-way heat exchanger to again provide sensible and latent cooling of outside air.

[0006] Three-way heat exchangers may include panels that channel the heat transfer fluid and the liquid desiccant therethrough for absorbing heat and moisture from the air stream that flows between the panels. The heat transfer fluid and liquid desiccant may flow through the panels and distribute across respective flow channels in each panel. There is an ongoing need for improvements in the design and/or manufacturability of the panels that facilitate reducing costs and/or optimizing operation and efficiency of the heat exchanger.

[0007] This background section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. These statements are to be read in this light, and not as admissions of prior art.

SUMMARY

[0008] In one aspect, a method of assembling a multilayer panel for use in a heat exchanger includes positioning a frame on a work platform, the frame having two header sections and a middle section, the middle section defining a heat transfer fluid area and each header section defining a header area. The method also includes positioning a heat exchange sheet on the frame across the middle section and each header section, welding the heat exchange sheet to the frame to form a middle weld enveloping the heat transfer fluid area and two outer welds each enveloping a corresponding one of the header areas, and controlling alignment between the frame and the heat exchange sheet during welding.

[0009] In another aspect, a method of assembling a multilayer panel for use in a heat exchanger includes positioning a frame on a work platform, the frame having two header sections and a middle section, the middle section defining a heat transfer fluid area and each header section defining a header area. The method also includes positioning a heat exchange sheet on the frame across the middle section and each header section, the heat exchange sheet having a sheet alignment feature including a flexure formed within the heat exchange sheet, the flexure defining a flexible region on the heat exchange sheet. The method also includes welding the heat exchange sheet to the frame to form a middle weld enveloping the heat transfer fluid area and two outer welds each enveloping a corresponding one of the header areas and controlling alignment between the frame and the heat exchanger sheet during welding using the flexure formed within the heat exchange sheet.

[0010] Various refinements exist of the features noted in relation to the above-mentioned aspects. Further features may also be incorporated in the above-mentioned aspects as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to any of the illustrated embodiments may be incorporated into any of the above-described aspects, alone or in any combination.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a schematic flow diagram of a heating, ventilation, and air conditioning (HVAC) system.

[0012] FIG. **2** is a front perspective of a three-way heat exchanger included in the HVAC system of FIG. **1**.

[0013] FIG. **3** is a front perspective of the three-way heat exchanger, with various components omitted to show internal components.

[0014] FIG. **4** is a rear perspective of the three-way heat exchanger.

[0015] FIG. **5** is a rear perspective of the three-way heat exchanger with various components omitted, similar to FIG. **3**.

[0016] FIG. **6** is a left side elevation of the three-way heat exchanger with various components omitted, similar to FIGS. **3** and **5**.

[0017] FIG. **7** is a right side elevation of an example panel assembly included in the three-way heat exchanger of FIGS. **2-6**.

[0018] FIG. **8** is an exploded view of the panel assembly of FIG. **7**.

[0019] FIG. **9** is a schematic section of the panel assembly taken along section line **9-9** in FIG. **7**.
[0020] FIGS. **10**A-**10**F are enlarged views of the sections A, B, C, D, E, and F, respectively, shown in FIG. **8**.
[0021] FIG. **11** is a schematic showing flow of liquid desiccant and heat transfer fluid through the three-way heat exchanger of FIGS. **2-6**.
[0022] FIG. **12** is a schematic of an example system that may be used to assemble the panel assembly of FIG. **7**.
[0023] FIGS. **13-15** schematically depict a sequence of operations for assembling the panel assembly of FIG. **7**.
[0024] FIG. **16** is an example method of assembling the panel assembly of FIG. **7** in accordance with the sequence of operations of FIGS. **13-15**.
[0025] Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

[0026] FIG. **1** is a schematic diagram of a heating, ventilation, and air conditioning (HVAC) system **100**. The HVAC system **100** includes sub-systems **102-106** and a liquid desiccant circuit **108** which facilitate the heating, cooling, and moisture removal capabilities of the system **100**. The sub-systems of the HVAC system **100** include a refrigerant sub-system **102**, a conditioner sub-system **104**, and a regenerator sub-system **106**. The conditioner sub-system **104** and the regenerator sub-system **106** are usable to respectively treat first and second inlet air streams **110** and **114**, and may be referred to herein as air treatment sub- systems **104** and **106**. The HVAC system **100** may include additional components or other components than those shown and described with reference to FIG. **1**.
[0027] In an example operating mode of the HVAC system **100**, the conditioner sub-system **104** removes heat from the first inlet air stream **110** and channels a conditioned outlet air stream **112** to a conditioned space (not shown), such as an interior of a building structure or vehicle. The conditioned outlet air stream **112** exiting the conditioner sub-system **104** may have a lower temperature than the first inlet air stream **110**. Heat removed from the first inlet air stream **110** is transferred from the conditioning sub-system **104**, to the refrigerant sub-system **102**, and finally to the regenerator sub-system **106**. The regenerator sub-system **106** transfers the heat into the second inlet air stream **114** and channels a heated outlet air stream **116** to the atmosphere.
[0028] The refrigerant sub-system **102** includes an evaporator **118**, a condenser **120**, a compressor **122**, and an expansion valve **124**. The compressor **122** may be any suitable compressor including, but not limited to, scroll, reciprocating, rotary, screw, and centrifugal compressors. The expansion valve **124** may be any suitable expansion valve, such as a thermal expansion valve. The expansion valve **124** may alternatively be any suitable expansion device, such as an orifice or capillary tube for example. The refrigerant sub-system **102** also includes a refrigerant loop **126** that circulates a working fluid, such as a refrigerant, between the evaporator **118**, the compressor **122**, the condenser **120**, and the expansion valve **124**. The refrigerant sub-system **102** may include additional components or other components than those shown and described with reference to FIG. **1**.
[0029] In an example operation of the refrigerant sub-system **102**, the refrigerant in the loop **126** is channeled as a low pressure gas refrigerant **128** toward the compressor **122**. The compressor **122** compresses the gas refrigerant **128**, which raises the temperature and pressure of the refrigerant. Pressurized, high temperature gas refrigerant **130** exits the compressor **122** and is channeled toward the condenser **120**, where the high pressure gas refrigerant **130** is condensed to a high pressure liquid refrigerant **132**. The liquid refrigerant **132** exiting the condenser **120** is channeled toward the expansion valve **124** that reduces the pressure of the liquid. The reduced pressure fluid refrigerant **134**, which may be a gas or a mixture of gas and liquid after passing through the expansion valve **124**, is then channeled toward the evaporator **118**. The fluid refrigerant **134** evaporates to a gas in the evaporator **118**, exiting the evaporator as the low pressure gas refrigerant **128**. The gas refrigerant **128** is then channeled back toward the compressor **122**, where the gas refrigerant **128** is

again compressed and the process repeats. Circulation of the refrigerant in the loop **126** may be driven by the compressor **122**, and, more particularly, by a pressure differential that exists between the pressurized, high temperature gas refrigerant **130** exiting the compressor **122** and the low pressure gas refrigerant **128** entering the compressor **122**. The direction of flow of the refrigerant through the loop **126**, as shown in FIG. **1**, may be reversed to switch the heat transfer functions of the evaporator **118** and the condenser **120**, and enable the HVAC system **100** to operate in various operating modes.

[0030] The conditioner sub-system **104** includes a first three-way heat exchanger **136** and a conditioner heat transfer fluid loop **138** that circulates a conditioner heat transfer fluid (e.g., water, a glycol-based fluid, or any combination thereof) to and from the first three-way heat exchanger **136**. The conditioner sub-system **104** interfaces with the refrigerant sub-system **102** via the evaporator **118**. In particular, the evaporator **118** is included in the refrigerant loop **126** and the conditioner heat transfer loop **138**, and facilitates transfer of heat from the conditioner heat transfer fluid in the loop **138** into the fluid refrigerant **134** in the refrigerant loop **126**. The conditioner sub-system **104** may include additional components or other components than those shown and described with reference to FIG. **1**. For example, the conditioner sub-system **104** may include one or more pumps (not shown) for circulating the conditioner heat transfer fluid in the loop **138** between the first three-way heat exchanger **136** and the evaporator **118**. Suitable pumps that may be included in the conditioner sub-system **104** include, for example, centrifugal pumps, diaphragm pumps, positive displacement pumps, or any type of pump suitable for transferring liquid. The conditioner sub-system **104** may include additional heat transfer equipment that transfers heat from the conditioner heat transfer fluid into the atmosphere, or vice versa, depending on the operational requirements of the HVAC system **100** and other factors (e.g., a temperature and/or humidity of the first air inlet stream **110**).

[0031] In an example operation of the conditioner sub-system **104**, the conditioner heat transfer fluid in the loop **138** is channeled toward the evaporator **118**. The conditioner heat transfer fluid is cooled in the evaporator **118** as heat is transferred from the conditioner heat transfer fluid into the fluid refrigerant **134** in the loop **126** to produce the gas refrigerant **128**. Cooled conditioner heat transfer fluid **140** exiting the evaporator **118** is channeled toward and enters the first three-way heat exchanger **136**. The first inlet air stream **110** is also directed through the first three-way heat exchanger **136**. The first three-way heat exchanger **136** transfers heat from the first inlet air stream **110** into the conditioner heat transfer fluid **140**, thus heating the conditioner heat transfer fluid. The heated conditioner heat transfer fluid **142** exiting the first three-way heat exchanger **136** is channeled back toward the evaporator **118** and the process repeats.

[0032] The regenerator sub-system **106** includes a second three-way heat exchanger **144** and a regenerator heat transfer fluid loop **146** that circulates a regenerator heat transfer fluid (e.g., water, a glycol-based fluid, or any combination thereof) to and from the second three-way heat exchanger **144**. The regenerator sub-system **106** interfaces with the refrigerant sub-system **102** via the condenser **120**. In particular, the condenser **120** is included in the refrigerant loop **126** and the regenerator heat transfer loop **146**, and facilitates transfer of heat from the pressurized gas refrigerant **130** in the refrigerant loop **126** into the regenerator heat transfer fluid. The regenerator sub-system **106** may include additional components or other components than those shown and described with reference to FIG. **1**. For example, the regenerator sub-system **106** may include one or more pumps (not shown) for circulating the regenerator heat transfer fluid in the loop **146** between the three-way heat exchanger **144** and the condenser **120**. Suitable pumps that may be included in the regenerator sub-system **106** include, for example, centrifugal pumps, diaphragm pumps, positive displacement pumps, or any type of pump suitable for transferring liquid. The regenerator sub-system **106** may include additional heat transfer equipment that transfers heat from the atmosphere into the regenerator heat transfer fluid, or vice versa, depending on the operational requirements of the HVAC system **100** and other factors (e.g., a temperature and/or humidity of the first air inlet stream **110**).

[0033] In an example operation of the regenerator sub-system **106**, the regenerator heat transfer fluid in the loop **146** is channeled toward the condenser **120**. The regenerator heat transfer fluid is heated in the condenser as heat is transferred from the pressurized gas refrigerant **130** in the loop **126** into the regenerator heat transfer fluid to produce the liquid refrigerant **132**. Heated regenerator heat transfer fluid **148** exiting the condenser is channeled toward and enters the second three-way heat exchanger **144**. The second inlet air stream **114** is also directed through the second three-way heat exchanger **144**. The second three-way heat exchanger **144** transfers heat from the regenerator heat transfer fluid into the second inlet air stream **114**, thus cooling the regenerator heat transfer fluid. The heated outlet air stream **116** exiting the second three-way heat exchanger **144** has a greater temperature than the second inlet air stream **114**. The cooled regenerator heat transfer fluid **150** exiting the three-way heat exchanger **144** is channeled back toward the condenser **120** and the process repeats.

[0034] The HVAC system **100** also includes the liquid desiccant circuit **108** that operates in conjunction with the sub-systems **102-106** to facilitate conditioning the first inlet air stream **110** by latent and sensible cooling. The liquid desiccant circuit **108** includes a liquid desiccant that is channeled between the first and second three-way heat exchangers **136** and **144**. Suitable liquid desiccants that may be used in the liquid desiccant circuit **108** include, for example, desiccant salt solutions, such as solutions of water and lithium chloride (LiCl), lithium bromide (LiBr), calcium chloride (CaCl.sub.2), or any combination thereof, triethylene glycol, sodium hydroxide, sulfuric acid, and so-called ionic liquid desiccants, or organic salts that are liquid at room temperature and have organic cations and organic or inorganic anions.

[0035] The liquid desiccant circuit **108** may include one or more pumps (not shown) for channeling the liquid desiccant between the first three-way heat exchanger **136** and the second three-way heat exchanger **144**. Suitable pumps that may be included in the liquid desiccant circuit **108** include, for example, centrifugal pumps, diaphragm pumps, positive displacement pumps, or any type of pump suitable for transferring liquid. The liquid desiccant circuit **108** may include one or more pumps for transferring the liquid desiccant from the second heat exchanger **144** toward the first heat exchanger **136** and one or more pumps for transferring the diluted liquid desiccant **154** from the first heat exchanger **136** toward the second heat exchanger **144**.

[0036] Concentrated liquid desiccant **152** in the liquid desiccant circuit **108** is channeled toward the first three-way heat exchanger **136** of the conditioner sub-system **104**, where the concentrated liquid desiccant **152** removes moisture from the first inlet air stream **110**. The concentrated liquid desiccant **152** cooperates with the cooled conditioner heat transfer fluid **140** in the first three-way heat exchanger **136** to absorb heat and moisture from the first inlet air stream **110**. The conditioned outlet air stream **112** exiting the first three-way heat exchanger **136** may have a lower humidity and/or a lower temperature than the first inlet air stream **110**. The liquid desiccant, having absorbed moisture from the first inlet air stream **110**, exits the first three-way heat exchanger **136** as diluted liquid desiccant **154**.

[0037] The diluted liquid desiccant **154** is channeled toward the second three-way heat exchanger **144** of the regenerator sub-system **106**, where the diluted liquid desiccant **154** rejects moisture into the second inlet air stream **114**. The diluted liquid desiccant **154** cooperates with the heated regenerator heat transfer fluid **148** in the second three-way heat exchanger **144** to reject heat and moisture into the second inlet air stream **114**. The heated outlet air stream **116** exiting the second three-way heat exchanger **144** thus has a greater humidity as well as a higher temperature than the second inlet air stream **114**. The liquid desiccant, having rejected moisture into the second inlet air stream **114**, exits the regenerator sub-system **106** as concentrated liquid desiccant **152**. The concentrated liquid desiccant **152** exiting the second three-way heat exchanger **144** is channeled back toward the first three-way heat exchanger **136**, and the process repeats.

[0038] The liquid desiccant circuit **108** may also include a desiccant-desiccant heat exchanger **156** for transferring heat from the concentrated liquid desiccant **152** that has exited the second three-way heat exchanger **144** to the diluted liquid desiccant **154** that has exited the first three-way heat exchanger **136**. The desiccant-desiccant heat exchanger **156** may facilitate improving the functions of the liquid desiccant in the three-way heat exchangers **136** and **144**. For example, the desiccant-desiccant heat exchanger **156** may reduce a temperature of the concentrated liquid desiccant **152** to provide greater cooling and dehumidifying capabilities of the first three-way heat exchanger **136**. Additionally and/or alternatively, the desiccant- desiccant heat exchanger **156** may increase a temperature of the diluted liquid desiccant **154** to enable the diluted liquid desiccant **154** to desorb a greater amount of moisture in the second three-way heat exchanger **144**. The desiccant-desiccant heat exchanger **156** may be an inline heat exchanger or any suitable heat exchanger that facilitates direct heat transfer between the concentrated liquid desiccant **152** and the diluted liquid desiccant **154**. The desiccant-desiccant heat exchanger **156** may alternatively facilitate indirect heat exchange between the concentrated liquid desiccant **152** and the diluted liquid desiccant **154**, such as via a vapor compression heat pump. Auxiliary heating and cooling sources (e.g., heating and cooling fluid, such as water) may also be utilized, in addition to or in lieu of the heat exchanger **156**, to respectively heat the diluted liquid desiccant **154** and cool the concentrated liquid desiccant **152**. The liquid desiccant circuit **108** may include additional components or other components than those shown and described with reference to FIG. **1**.

[0039] Thus, in the example operating mode of the HVAC system **100**, sensible cooling of the first inlet air stream **110** is facilitated by the first three-way heat exchanger **136** of the conditioner sub-system **104**, which transfers heat from the inlet air stream **110** into the conditioner heat transfer fluid. The heat removed from the first inlet air stream **110** is then transferred sequentially between the sub-systems **104**, **102**, and **106** via the evaporator **118** and the condenser **120**, and eventually is rejected into the second inlet air stream **114** via the second three-way heat exchanger **144**. Latent cooling of the first inlet air stream **110** is also facilitated by the first three-way heat exchanger **136**, which removes moisture from the inlet air stream **110** using the concentrated liquid desiccant **152**. The moisture absorbed by the diluted liquid desiccant **154** is desorbed in the second three-way heat exchanger **144** into the second inlet air stream **114**, which regenerates the concentrated liquid desiccant **152** that is then channeled back toward the first three-way heat exchanger **136**.

[0040] The HVAC system **100** may operate in alternative operating modes than the example operating mode described above with reference to FIG. **1**. The example operating mode of the HVAC system **100** described above may be considered a warm weather operating mode of the HVAC system **100**, in which warm, humid air in the first inlet air stream **110** is cooled and dehumidified using the conditioner sub-system **104** and the heat and moisture removed is transferred by the sub-systems **102** and **106** and the liquid desiccant circuit **108** and rejected into the second inlet air stream **114** to produce the heated, humidified outlet air stream **116** that is directed into the warm, humid ambient. In a cold weather operating mode of the HVAC system **100**, the operation of the sub-systems **102-106** and the liquid desiccant circuit **108** may be reversed such that the first three-way heat exchanger **136** heats and humidifies cool, dry air in the first inlet air stream **110** to produce warm air with a comfortable humidity level in the outlet air stream **112** that is channeled to a conditioned space. In the cold weather operating mode, the direction of flow of the refrigerant in the loop **126** and the liquid desiccant in the liquid desiccant circuit **108** may be reversed, such that the air treatment sub-systems **104** and **106** switch their respective functions, or intake and outlet vents for the first and second inlet air streams **110** and **114** may be rearranged and/or reconfigured such that the direction of airflow directed through the first and second three-way heat exchangers **136** and **144** is reversed, with the outlet air stream **112** being channeled toward the ambient environment and the outlet air stream **116** being channeled toward the conditioned space. In still other operating modes of the HVAC system **100**, depending on the operational requirements and desired setpoint temperature and humidity level within the conditioned space, one of the air treatment sub-system **104** and **106** may be idle or omitted from the HVAC system **100**. For example, the air treatment sub-system **106** may be omitted and the refrigerant sub-system **104** may reject or absorb heat from a refrigerant-air heat exchanger **120**, depending on the operating mode of the HVAC system **100**. Where the regenerator sub-system **106** is omitted or idle, liquid desiccant in the liquid desiccant circuit **108** that is cycled through the first three-way heat exchanger **136** may be regenerated or diluted, depending on the operating mode of the HVAC system **100**, using auxiliary regeneration equipment, dilution tanks, or the like.

[0041] Still with reference to FIG. **1**, the first three-way heat exchanger **136** and the second three-way heat exchanger **144** have substantially the same configuration. In alternative embodiments, the first three-way heat exchanger **136** and the second three-way heat exchanger **144** may have a different configuration. Although the conditioner sub-system **104** and the regenerator sub-system **106** are shown in FIG. **1** to include one three-way heat exchanger **136** and **144**, respectively, any suitable number of three-way heat exchangers **136** and **144** may be included in the respective sub-system **104** and **106**. The number of three-way heat exchangers **136** included in the conditioner sub-system **104** may be the same or different than the number of three-way heat exchangers **144** included in the regenerator sub-system **106**. Where the conditioner sub-system **104** includes multiple three-way heat exchangers **136**, the heat exchangers **136** may operate in series, in parallel, or any combination thereof. Where the regenerator sub-system **106** includes multiple three-way heat exchangers **144**, the heat exchangers **144** may operate in series, in parallel, or any combination thereof.

[0042] Referring now to FIGS. **2-5**, an example three-way heat exchanger **200** for use in an air treatment sub-system of the HVAC system **100** of FIG. **1** will now be described. The three-way heat exchanger **200** may be implemented as a first three-way heat exchanger **136** in the conditioner sub-system **104** and/or as a second three-way heat exchanger **144** in the regenerator sub-system **106**. FIG. **2** is a front perspective of the three-way heat exchanger **200**. FIG. **3** is a front perspective of the three-way heat exchanger **200** with various components omitted to show internal components of the three-way heat exchanger **200**. FIG. **4** is a rear perspective of the three-way heat exchanger **200**. FIG. **5** is a rear perspective of the three-way heat exchanger **200** with various components omitted, similar to FIG. **3**.

[0043] The three-way heat exchanger **200** has a dimension in the X-axis, Y-axis and Z-axis, respectively. The X-axis, Y-axis and Z-axis are each mutually perpendicular. As described herein with respect to the three-way heat exchanger **200** and components of the heat exchanger **200** when assembled, dimensions in the Z-axis may be referred to as a “height,” dimensions in the Y-axis may be referred to as a “length,” and dimensions in the X-axis may be referred to as a “width.” The three-way heat exchanger **200** defines a lateral direction in the X-axis, a longitudinal direction in the Y-axis, and a vertical direction in the Z-axis. The X-axis may also be referred to herein as a lateral axis, the Y-axis may also be referred to herein as a longitudinal axis, and the Z-axis may also be referred to herein as a vertical axis. The three-way heat exchanger **200** has respectively opposite first and second lateral sides **202** and **204**, first and second longitudinal sides **206** and **208**, and first and second vertical sides **210** and **212**. The first and second lateral sides **202** and **204** are spaced apart in the lateral direction, the first and second longitudinal sides **206** and **208** are spaced apart in the longitudinal direction, and the first and second vertical sides **210** and **212** are spaced apart in the vertical direction. Directional terms are used for solely for description of the three-way heat exchanger **200** and the spatial relation of the components of the heat exchanger. The examples shown and described are not limited to any particular orientation.

[0044] The three-way heat exchanger **200** includes a set of panel assemblies **214** (also referred to as multilayer panels) arranged in succession or series in the lateral direction between the first lateral side **202** and the second lateral side **204**. The individual panel assemblies **214** will be described in more detail below with reference to FIGS. **7-10**. Each panel assembly **214** is in the form of a plate structure that has an internal heat transfer fluid channel through which a heat transfer fluid, such as the conditioner heat transfer fluid in the loop **138** or regenerator heat transfer fluid in the loop **146**, flows. Each panel assembly **214** also includes liquid desiccant channels on opposite sides of the heat transfer fluid channel. A liquid desiccant, such as the concentrated liquid desiccant **152** or the diluted liquid desiccant **154** in the liquid desiccant circuit **108**, flows through the liquid desiccant channels. Liquid desiccant flowing through the liquid desiccant channels is separated from the heat transfer fluid flowing through the heat transfer fluid channel of the respective panel assembly, and heat is exchanged between the liquid desiccant in the liquid desiccant channels and the heat transfer fluid flowing through the heat transfer fluid channel. Airflow gaps **216**, also referred to as air gaps **216**, are defined between adjacent panel assemblies **214** in the lateral direction. Each airflow gap **216** extends primarily in the vertical and longitudinal directions.

[0045] Any suitable number of panel assemblies **214** may be included in the three-way heat exchanger **200**. For example, the three-way heat exchanger **200** may include from 1 to 200 panel assemblies **214**, from 1 to 100 panel assemblies **214**, from 50 to 200 panel assemblies **214**, from 50 to 100 panel assemblies **214**, such as one panel assembly, ten panel assemblies **214**, twenty panel assemblies **214**, thirty panel assemblies **214**, forty panel assemblies **214**, fifty panel assemblies **214**, sixty panel assemblies **214**, seventy panel assemblies **214**, eighty panel assemblies **214**, ninety panel assemblies **214**, 100 panel assemblies **214**, or greater than 100 panel assemblies **214**.

[0046] The panel assemblies **214** are supported on a base **240** at the second vertical side **212** of the three-way heat exchanger **200**. The panel assemblies **214** extend substantially parallel to one another between the base **240** and the first vertical side **210** of the three-way heat exchanger **200**. The panel assemblies **214** may, in an example operation of the three-way heat exchanger **200**, deviate from a substantially parallel extent as fluid flows through the panel assemblies **214** and/or as air flows through the air gaps **216** between adjacent panel assemblies **214**. The base **240** includes a liquid desiccant reservoir (shown in FIGS. **4** and **5**) adjacent the airflow outlet **226** at the second longitudinal side **208** and the second vertical side **212**. The liquid desiccant reservoir extends longitudinally outward beyond the panel assemblies **214** and may collect liquid desiccant that is entrained in and subsequently removed from the air stream at the airflow outlet **226** using a liquid desiccant mist trap, described, for example, in U.S. patent application Ser. No. 18/391,384, titled “LIQUID DESICCANT AIR CONDITIONER MODULES HAVING A LIQUID DESICCANT MIST TRAP,” filed Dec. 20, 2023, the disclosure of which is incorporated by reference in its entirety.

[0047] The three-way heat exchanger **200** includes a first end plate **218** and a second end plate **220** at the first and second lateral sides **202** and **204**, respectively. The end plates **218**, **220** may also be referred to as end covers or end sheets. The end plates **218** and **220** may provide lateral support for the set of panel assemblies **214** and enclose an interior **222** of the three-way heat exchanger **200** at the first and second lateral sides **202** and **204**. The end plates **218** and **220** are omitted from FIGS. **3** and **5** to show the arrangement of the panel assemblies **214**, the airflow gaps **216** defined between adjacent panel assemblies **214**, and the interior **222** of the three-way heat exchanger **200** in greater detail.

[0048] Each end plate **218** and **220** includes alignment apertures **258** and **260**, respectively, for receiving clamping assemblies (not shown) used to clamp the panel assemblies **214** together. Example clamping assemblies suitable for use in the three-way heat exchanger **200** are described in U.S. patent application Ser. No. 18/490,984, filed Oct. 20, 2023, the disclosure of which is incorporated by reference in its entirety.

[0049] The interior **222** of the three-way heat exchanger **200** may be enclosed at the first and second vertical sides **210** and **212** of the three-way heat exchanger by the set of panel assemblies **214**. For example, adjacent panel assemblies **214** may be connected and/or in contact with one another at opposite vertical ends to seal the respective airflow gap **216** defined therebetween at the opposite vertical ends and to enclose the interior **222** of the three-way heat exchanger at the first and second vertical sides **210** and **212**. Additionally and/or alternatively, the three-way heat exchanger **200** may include vertical end plates (not shown) to enclose the interior **222** at the first and second vertical sides **210** and **212**.

[0050] The three-way heat exchanger **200** includes an airflow inlet **224** on the first longitudinal side **206** and an airflow outlet **226** on the second longitudinal side **208**. The airflow inlet **224** and the airflow outlet **226** are respectively defined by longitudinal side panels **228** and **230** of the three-way heat exchanger **200**. For example, the longitudinal side panels **228** and **230** may include openings in the form of grated or grille openings, shutters, louvers, dampers, or may have any other suitable open configuration to enable airflow to enter into and exit the three-way heat exchanger **200**. In some examples, one or both of the longitudinal side panels **228** and **230** may include a filter to filter particulate and/or contaminants from an air stream that is treated by the three-way heat exchanger **200**. The airflow inlet **224** and the airflow outlet **226** are in communication with the airflow gaps **216** defined between the adjacent panel assemblies **214**, and allow an inlet air stream (e.g., the first or second inlet air stream **110** or **114** in FIG. **1**) to flow in an airflow direction (indicated by the arrow **278** in FIG. **2**) through the three-way heat exchanger **200** in the longitudinal direction. The airflow direction **278** is in the longitudinal direction in the illustrated example. The airflow direction **278** may be additionally and/or alternatively be in the lateral and/or vertical directions. The longitudinal side panels **228** and **230** are omitted from FIGS. **3** and **5** to show the arrangement of the panel assemblies **214**, the airflow gaps **216** defined between adjacent panel assemblies **214**, and the interior **222** of the three-way heat exchanger **200** in greater detail.

[0051] The three-way heat exchanger **200** also includes heat transfer fluid inlet **232** and outlet **234** and liquid desiccant inlet **236** and outlet **238**. Heat transfer fluid (e.g., circulating in one of the heat transfer fluid loops **138** or **146** in FIG. **1**) enters into and exits the three-way heat exchanger **200** via the heat transfer fluid inlet **232** and outlet **234**, respectively. Liquid desiccant (e.g., circulating the liquid desiccant circuit **108** in FIG. **1**) enters into and exits the three-way heat exchanger **200** via the liquid desiccant inlet **236** and outlet **238**, respectively. The locations of the heat transfer fluid inlet **232** and outlet **234** and the liquid desiccant inlet **236** and outlet **238** may vary depending on a desired flow direction of the heat transfer fluid and the liquid desiccant through the panel assemblies **214**. The liquid desiccant inlet **236** and the heat transfer fluid outlet **234** may be defined by (e.g., made integral with) the end plate **218** and the liquid desiccant outlet **238** and the heat transfer fluid inlet **232** may be defined by (e.g., made integral with) the end plate **220**. Alternatively, the heat transfer fluid inlet **232** and outlet **234** and the liquid desiccant inlet **236** and outlet **238** may each be defined by a conduit (e.g., a pipe, tube, hose, or other suitable fluid conduit) that extends longitudinally through an opening in the respective end plate **218** and **220**.

[0052] Referring to FIGS. **7-9**, an example panel assembly **300** (also referred to as a multilayer panel) suitable for use as the individual panel assemblies **214** will now be described. In the example three-way heat exchanger **200**, all the panel assemblies **214** have substantially the same configuration as the panel assembly **300** shown in FIGS. **7-9**. The panel assemblies **214** will be referred to as the panel assemblies **300** hereinafter for convenience of description. Some of or all the panel assemblies **214** may include additional components, fewer components, or other components than the panel assembly **300**.

[0053] FIG. **7** is a right side elevation of the example panel assembly **300**. FIG. **8** is an exploded view of the panel assembly **300**. FIG. **9** is a schematic section of the panel assembly **300** taken along section line **9-9** in FIG. **7**. The spatial relation of components of the panel assembly **300** will be described with respect to the X-axis, Y-axis, and Z-axis, and the lateral direction, longitudinal direction, and vertical direction defined by the three-way heat exchanger **200**. The panel assembly **300** will also be described in the orientation when implemented and installed in the three-way heat exchanger **200**. Directional terms are used to describe components of the panel assembly **300** solely for convenience of description. The examples shown and described are not limited to any particular orientation.

[0054] The panel assembly **300** includes a frame **302** that defines first and second vertical ends **304** and **306**, in the Z-axis, first and second lateral faces **305** and **307**, in the X-axis, and first and second longitudinal ends **308** and **310**, in the Y-axis, of the panel assembly **300**. The frame **302** includes opposite first and second header sections **312** and **314** respectively located at the first and second vertical ends **304** and **306**. The frame **302** also includes a middle section **316** between the opposite header sections **312** and **314**. The header sections **312** and **314** respectively define liquid desiccant header areas **320** and **322**. The middle section **316** defines a heat transfer fluid area **324**. The liquid desiccant header areas **320** and **322** are separated from the heat transfer fluid area **324** by portions **325**, **327** of the frame **302** (or "frame bars" **325**, **327**) that respectively extend between the heat transfer fluid area **324** and one of the liquid desiccant header areas **320** and **322**. The middle section **316** defines a leading edge and a trailing edge of the frame **302**. The leading edge extends between the header sections **312**, **314** proximate the first longitudinal end **308**, and the trailing edge extends between the header sections **312**, **314** proximate the second longitudinal end **310**. The leading edge and/or the trailing edge may include acrodynamic features that facilitate controlling a pressure drop and/or reducing drag of the air flowing through the airflow gaps **216**. Example acrodynamic features are described in U.S. patent application Ser. No. 18/390,941, titled "LIQUID DESICCANT AIR CONDITIONER MODULES HAVING AERODYNAMIC FEATURES," filed Dec. 20, 2023, the disclosure of which is incorporated by reference in its entirety. Each header section **312**, **314** of the frame **302** includes complementing airflow restrictor members (not labeled) that cooperate or engage with the airflow restrictor members of the adjacent panel assembly **300** when the heat exchanger **200** is assembled to form airflow restrictors located in the airflow gap **216** between the adjacent panel assemblies at opposite vertical ends. The airflow restrictors are described in U.S. patent application Ser. No. 18/390,648, titled "LIQUID DESICCANT AIR CONDITIONER MODULES HAVING INTERLOCKING PANELS FOR CONTROLLING AIRFLOW," filed Dec. 20, 2023, the disclosure of which is incorporated by reference in its entirety.

[0055] The panel assembly **300** also includes first and second plates **326** and **328** disposed on opposite lateral faces of the frame **302**, covering the middle section **316** of the frame **302**. The first and second plates **326** and **328** be a sheet of material that is, for example, less than 0.5 inch thick, or less than 0.25 inch thick, and the plates **326**, **328** may also be referred to as "heat exchange sheets" or "sheets." The first and second plates **326** and **328** may be attached to the frame **302** or may be made integral with the frame **302**. Suitable techniques for attaching the plates **326** and **328** to the frame **302** may include, for example, welding (e.g., laser, induction, or radio-frequency welding), adhesive bonding, thermal bonding, or another suitable technique for joining materials together. Additional detail on attaching the plates **326** and **328** to the frame **302** is described in U.S. Pat. No. 11,022,330, issued on Jun. 1, 2021, and U.S. Pat. No. 10,921,001, issued on Feb. 16, 2021, the disclosures of each of which are hereby incorporated herein by reference in their entirety. Example systems and methods for attaching the plates **326** and **328** to the frame **302** are described further below with reference to FIGS. **12-16**.

[0056] The frame **302** and the plates **326** and **328** may be made from dissimilar but compatible materials for welding together. For example, the frame **302** and the plates **326** and **328** may each be made from the same or different thermoplastic or polymer materials. The material used for the frame **302** and the plates **326** and **328** may also be selected based on its compatibility with the liquid desiccant used in the three-way heat exchanger **200**. Suitable polymer materials for the frame **302** and the plates **326** and **328** include, for example, polyolefins (e.g., polypropylene and/or polyethylene), acrylonitrile butadiene styrene (ABS), and combinations thereof. The frame **302** and the plates **326**, **328** may be made from the same or compatible materials to enable attaching (e.g., welding) the frame and the plates. The frame **302** and/or the plates **326** and **328** may include additives that improve properties such as laser-absorbing and conductivity properties, as well as the strength and/or stiffness of the plates **326** and **328**. For example, the frame **302** and/or the plates **326**, **328** may include carbon black in a suitable amount to facilitate absorbing laser energy and laser welding the plates and the frame. Additionally, and/or alternatively, the frame **302** and/or the plates **326**, **328** may include conductive fiber additives to facilitate radio-frequency welding. In other examples, the frame **302** and the plates **326** and **328** may be made from any other suitable materials that enable the three-way heat exchanger **200** to function as described.

[0057] The sheets or plates **326** and **328** envelop and seal the heat transfer fluid area **324** of the frame, defining a heat transfer fluid channel **330** of the panel assembly **300** between the plates **326** and **328** (see FIG. **9**). As described below, in an example operation of the three-way heat exchanger **200**, heat transfer fluid flows between the plates **326** and **328** through the heat transfer fluid channel **330** and liquid desiccant flows over an outer surface of the plates **326** and **328**, opposite the heat transfer fluid channel **330**. The plates **326** and **328** isolate the liquid desiccant from the heat transfer fluid in the channel **330**, and allow heat to transfer between the liquid desiccant and the heat transfer fluid. The plates **326** and **328** may extend over one or both of liquid desiccant header areas **320** and **322** and define openings (e.g., apertures **360**) that align with the one or both of the liquid desiccant header areas **320** and **322** to enable liquid desiccant to flow therethrough. In the example panel assembly **300**, each of the plates **326** and **328** includes a series of apertures **360** located adjacent the liquid desiccant header **320** and a series of apertures **362** located adjacent the liquid desiccant header area **322**. Liquid desiccant may flow through the apertures **360** and **362** of each plate **326** and **328** to enter and/or exit the liquid desiccant header area **320** and **322**, respectively.

[0058] A netting or mesh (not shown) may be disposed in the heat transfer fluid channel **330** to maintain a width of the heat transfer fluid channel under negative pressure. The netting or mesh may also facilitate more constant flow rates of the heat transfer fluid through the channel **330**. The netting or mesh may also facilitate improving flow distribution of the heat transfer fluid between the panel assemblies **300** in the three-way heat exchanger **200**. The netting or mesh may also provide turbulation of the heat transfer fluid to increase heat transfer with the liquid desiccant flowing over the outer surfaces of the sheets or plates **326** and **328**. A wide variety of materials may be used for the netting or mesh. For example, the netting or mesh may include the same polymer material as the plates (e.g., polyolefins, ABS, or combinations thereof). Alternatively, a flow guide (not shown) may be disposed in the heat transfer fluid channel **330**. Example flow guides are described in U.S. patent application Ser. No. 18/585,344, titled “THREE-WAY HEAT EXCHANGE MODULE WITH CONTROLLED FLUID FLOW,” filed Feb. 23, 2024, the disclosure of which is incorporated by reference in its entirety.

[0059] Referring again to FIGS. **7-9**, the panel assembly **300** also includes membranes **332** and **334** disposed on the opposite lateral faces **305** and **307** of the frame **302**. In other examples, only one of the membranes **332** or **334** may be included in the panel assembly **300**. The membranes **332** and **334** cover the outer surfaces of the sheets or plates **326** and **328**. As shown in FIG. **9**, liquid desiccant channels **336** and **338** are respectively defined between the membrane **332** and the plate **326** and the membrane **334** and the plate **328**. The membranes **332** and **334** also envelop and seal the liquid desiccant header areas **320** and **322**. Each liquid desiccant channel **336** and **338** connects the liquid desiccant header areas **320** and **322** in fluid communication. As described below, in an example operation of the three-way heat exchanger **200**, liquid desiccant flows through one of the liquid desiccant header areas **320** or **322**, into the liquid desiccant channels **336** and **338**, over the outer surfaces of the plates **326** and **328** and behind the membranes **332** and **334**, and finally into the other one of the liquid desiccant header areas **320** or **322**. The plates **326** and **328** limit contact between the liquid desiccant flowing in the liquid desiccant channels **336** and **338** and heat transfer fluid flowing through the heat transfer fluid channel **330**, and enable heat to transfer therebetween. In examples where only one of the membranes **332** or **334** is included in the panel assembly **300**, only one liquid desiccant channel **336** or **338** may be defined between the membrane **332** or **334** and the plate **326** or **328**. In these examples, the plate **326** or **328** on the lateral face **305** or **307** opposite the liquid desiccant channel **336** or **338** may envelop and seal the liquid desiccant header areas **320** and **322** and restrict flow of liquid desiccant opposite the liquid desiccant channel **336** or

338.

[0060] The membranes **332** and **334** are attached to one of the lateral faces **305** and **307**, respectively, of the frame **302** to envelop and seal the liquid desiccant header areas **320** and **322**. The membranes **332** and **334** may additionally and/or alternatively be attached to the outer surface of the respective sheet or plate **326** and **328**, which may facilitate maintaining a width of the liquid desiccant channels **336** and **338** and/or limiting a propensity of the membranes **332** and **334** to bulge outward when liquid desiccant flows through the liquid desiccant channels **336** and **338**. The membranes **332** and **334** may be attached to the lateral faces **305** and **307** of the frame **302** and/or the outer surfaces of the plates **326** and **328** using any suitable technique, such as adhesive bonding, heat sealing, or welding, for example. The membranes **332** and **334** may be respectively attached to the plates **326** and **328** directly by heat scaling or welding where compatible materials (e.g., polyolefins) are used for the membranes **332** and **334** and the respective plates **326** and **328**. An outer bonding layer (not shown) may be applied over the outer surfaces of the plates **326** and **328** to improve the quality or case of forming the heat seals or welds with the respective membranes **332** and **334**. The outer surfaces of the plates **326** and **328** may include raised patterns or dot features (not shown) to which the membranes **332** and **334** are adhered, heat sealed, or otherwise attached. The raised patterns may be formed on the frame **302** and/or plates **326** and **328** by thermoforming, embossing, or other suitable techniques. Attaching the membranes **332** and **334** to the dot features or raised patterns may provide an additional advantage of facilitating uniform distribution of the liquid desiccant across the liquid desiccant channels **336** and **338** in the longitudinal direction and reducing stresses that can lead to warping of the plates **326** and **328**. Warping of the plates **326** and **328** may degrade the ability to transfer heat and moisture between the heat transfer fluid, the liquid desiccant, and air that flows across the membranes **332** and **334** in an example operation of the three-way heat exchanger **200**. Additional detail on attaching the membranes **332** and **334** to the frame **302** and the respective plates **326** and **328** is described in U.S. Pat. No. 11,022,330, issued on Jun. 1, 2021, and U.S. Pat. No. 10,921,001, issued on Feb. 16, 2021, the disclosures of each of which are hereby incorporated herein by reference in their entirety. Example systems and methods for attaching the membranes **332** and **334** to the frame **302** and the respective sheets or plates **326** and **328** are described further below with reference to FIGS. **12-16**.

[0061] The membranes **332** and **334** are made of a vapor-permeable material that permits transfer of water vapor therethrough to enable liquid desiccant flowing in the liquid desiccant channels **336** and **338** to absorb moisture from and desorb moisture into air flowing across the membranes **332** and **334**. In some examples, the membranes **332** and **334** may each be made from a polypropylene material or other suitable vapor-permeable polymer material. The vapor-permeable material used for the membranes **332** and **334** may be microporous (e.g., having a pore size less than 0.5 micrometers (μm)). Examples of suitable microporous membranes are disclosed in U.S. Pat. No. 9,101,874, issued on Aug. 11, 2015, the disclosure of which is hereby incorporated by reference herein in its entirety. By way of example, suitable commercially available membrane include the EZ2090 polypropylene, microporous membrane from Celgard. Microporous membranes **332** and **334** may have 40-80% open area, pore sizes of less than 0.5 μm, and a thickness of less than 100 μm. Some example microporous membranes may have greater than 80% open area. One suitable membrane is approximately 65% open area and has a thickness of approximate 20 μm. This type of membrane is structurally very uniform in pore size and is thin enough to not create a significant thermal barrier. Other possible membranes include membranes from 3M, Lydall, and other manufacturers. The membranes **332** and **334** may include any suitable vapor-permeable material that permits water transfer therethrough to enable the liquid desiccant in the liquid desiccant channels **336** and **338** to absorb moisture from or desorb moisture into air flowing over the membranes **332** and **334**. The membranes **332** and **334** may be made from a translucent material, such as a translucent thermoplastic or polymer (e.g., polypropylene), that has a suitable laser transmissivity to enable laser welding the frame **302** and the plate **326**, **328** when the respective membrane **332**, **334** is attached (e.g., heat sealed) to the plate, without forming a weld between the membrane and plate.

[0062] The frame **302** defines a liquid desiccant inlet port **340** that feeds liquid desiccant into the liquid desiccant header area **320** and a liquid desiccant outlet port **342** that receives liquid desiccant from the liquid desiccant header area **322**. The liquid desiccant inlet port **340** is defined in a first corner flange **364** of the frame **302**. The first corner flange **364** of the frame **302** is part of the first header section **312** and is located adjacent to the liquid desiccant header area **320** at the first vertical end **304** and the second longitudinal end **310** of the panel assembly **300**. The liquid desiccant outlet port **342** is defined in a second corner flange **366** of the frame **302**. The second corner flange **366** is part of the second header section **314**, and is located adjacent to the liquid desiccant header area **322** at the second vertical end **306** and the first longitudinal end **308** of the panel assembly **300**. Thus, the first and second corner flanges **364** and **366**, and the liquid desiccant inlet and outlet ports **340** and **342** respectively defined therein, are located on opposite longitudinal and vertical ends of the panel assembly **300**.

[0063] As represented by the flow lines **344** in FIGS. **7** and **9**, in an example operation of the three-way heat exchanger **200**, liquid desiccant is supplied into the liquid desiccant header area **320** of the panel assembly **300** via the liquid desiccant inlet port **340**, flows through each of the liquid desiccant channels **336** and **338** and into the liquid desiccant header area **322**, and exits the panel assembly **300** via the liquid desiccant outlet port **342**. In the illustrated example, the liquid desiccant flows vertically downward in the desiccant channels **336**, **338**. The liquid desiccant may have alternative flow directions. The flow direction(s) of the liquid desiccant in the channels **336**, **338** may vary depending, for example, on the orientation of the panel assemblies **300** in the heat exchanger **200**, location of the liquid desiccant inlet and outlet ports **340**, **342**, and/or to which liquid desiccant header area **320**, **322** the liquid desiccant is supplied and from which header area the liquid desiccant exits.

[0064] The frame **302** also defines a heat transfer fluid inlet port **346** that feeds heat transfer fluid into the heat transfer fluid channel **330** and a heat transfer fluid outlet port **348** that receives heat transfer fluid from the heat transfer fluid channel **330**. The heat transfer fluid inlet port **346** is defined in a third corner flange **368** of the frame **302**. The third corner flange **368** of the frame **302** is part of the second header section **314** and the middle section **316**. The third corner flange **368** is located adjacent to heat transfer fluid channel **330**, at the second longitudinal end **310** and proximate to the second vertical end **306** of the panel assembly **300**. The heat transfer fluid outlet port **348** is defined in a fourth corner flange **370** of the frame **302**. The fourth corner flange **370** is part of the first header section **312** and the middle section **316**. The fourth corner flange **370** is located adjacent to the heat transfer fluid channel **330**, at the first longitudinal end **308** and proximate to the first vertical end **304** of the panel assembly **300**. Thus, the third and fourth corner flanges **368** and **370**, and the heat transfer fluid inlet and outlet ports **346** and **348** respectively defined therein, are located on opposite longitudinal and vertical ends of the panel assembly **300**. Additionally, the first and fourth corner flanges **364** and **370**, and the liquid desiccant inlet port **340** and the heat transfer fluid outlet port **348** respectively defined therein, are both located proximate to the first vertical end **304**, on opposite longitudinal ends of the panel assembly. The second and third corner flanges **366** and **368**, and the liquid desiccant outlet port **342** and the heat transfer fluid inlet port **346** respectively defined therein, are both located proximate to the second vertical end **306**, on opposite longitudinal ends of the panel assembly.

[0065] As represented by the flow lines **350** in FIGS. **7** and **9**, in an example operation of the three-way heat exchanger **200**, heat transfer fluid is supplied into the heat transfer fluid channel **330** of the panel assembly **300** via the heat transfer fluid inlet port **346**, flows therethrough, and exits the panel assembly **300** via the heat transfer fluid outlet port **348**. In the illustrated example, the heat transfer fluid flows vertically upward in the channel **330**. The heat transfer fluid may have alternative flow directions. The flow direction(s) of the heat transfer fluid in the channel **330** may vary depending, for example, on the orientation of the panel assemblies **300** in the heat exchanger **200**, location of the heat transfer fluid inlet and outlet ports **346**, **348**, and/or through which port **346**, **348** the heat transfer fluid enters the channel **330** and through which port the heat transfer fluid exits the channel **330**.

[0066] FIGS. **10**A-**10**D are enlarged views of the sections A, B, C, D, respectively, of the frame **302** shown in FIG. **8**, and depict the corner flanges **364-370** in greater detail. In particular, FIGS. **10**A-**10**D show microchannels or apertures that provide fluid connection between the ports **340**, **342**, **346**, and **348** and the respective fluid areas defined in the panel assembly **300**. As shown in FIGS. **10**A and **10**B, the heat transfer fluid inlet port **346** is connected to the heat transfer fluid area **324** by apertures **352** (FIG. **10**B) and the heat transfer fluid outlet port **348** is connected to the heat transfer fluid area **324** by apertures **354** (FIG. **10**A). The heat transfer fluid area **324** defines the heat transfer fluid channel **330** when sealed on the opposite lateral faces **305** and **307** of the frame **302** by the plates **326** and **328**. As indicated by the flow lines **350** in FIGS. **10**A and **10**B, heat transfer fluid enters the heat transfer fluid channel **330** from the inlet port **346** via the apertures **352** and exits the channel **330** into the outlet port **348** via the apertures **354**.

[0067] As shown in FIGS. **10**C and **10**D, the liquid desiccant inlet port **340** is connected to the first liquid desiccant header area **320** by apertures **356** (FIG. **10**C) and the liquid desiccant outlet port **342** is connected to the second liquid desiccant header area **322** by apertures **358** (FIG. **10**D). As indicated by the flow lines **344** in FIGS. **10**C and **10**D, liquid desiccant enters the first liquid desiccant header area **320** from the inlet port **340** via the apertures **356**, flows into and through the liquid desiccant channels **336** and **338** (FIG. **9**), into the second liquid desiccant header area **322**, and exits the header area **322** into the outlet port **342** via the apertures **358**. In the illustrated example of FIGS. **10**A-**10**D, each of the apertures **352-358** include two apertures. Any suitable number of apertures may be used for the apertures **352-358**. In some examples, a greater number of apertures may be used for some of the apertures **352-358** than the other apertures **352-358**. The number, size, and/or shape of the apertures **352-358** may be the same or different. The number of apertures, as well as the size and shape, used for each of the apertures **352-358** may also vary between panel assemblies **300**.

[0068] FIGS. **10**A-**10**D also show example features of the panel assembly **300** that may facilitate connecting adjacent panel assemblies **300** together when installed in the three-way heat exchanger **200**. For example, when panel assemblies **300** are arranged in series and installed in the heat exchanger **200**, the corner flanges **364-370** of the frames **302** of adjacent panel assemblies **300** may be connected. As shown, each corner flange **364-370** includes one or more snap fittings **372**. The snap fittings **372** extend in the lateral direction from the first lateral face **305** of the frame **302**, and a corresponding bore **374** (shown in FIG. **6**) depends into the second lateral face **307** at the location laterally opposite the snap fitting **372**. Each corner flange **364-370** includes two snap fittings **372** and corresponding bores **374** in the illustrated example. In other examples, more or fewer snap fittings **372** and corresponding bores **374** may be included at the corner flanges **364-370**. The corner flanges **364-370** may include the same or a different number of snap fittings **372** and corresponding bores **374**. Suitably, a corresponding bore **374** is included for each snap fitting **372** in each corner flange **364-370**. When the panel assemblies **300** are arranged in series and installed in the heat exchanger **200**, each snap fitting **372** of the corner flanges **364-370** of the frame **302** of one of the panel assemblies **300** is received by one of the corresponding bores **374** of the corner flanges **364-370** of the frame **302** of a laterally adjacent panel assembly **300** to directly connect adjacent panel assemblies **300**.

[0069] Each corner flange **364-370** also includes one or more alignment holes **376** extending therethrough in the lateral direction. In the illustrated embodiment, the corner flanges **364-370** each include two alignment holes **376**, labeled in FIGS. **10**A-D as a first alignment hole **376***a* and a second alignment hole **376***b*. The first alignment holes **376***a* of each corner flange **364-370** are located longitudinally inboard of the second alignment holes **376***b* and adjacent to respective corners of the heat transfer fluid area **324**. In other examples, more or fewer alignment holes **376** may be included at the corner flanges **364-370**. The corner flanges **364-370** may include the same or a different number of alignment holes **376**. When the panel assemblies are arranged in series and installed in the heat exchanger **200**, the alignment holes **376** of the corner flanges **364-370** of the frames **302** of the panel assemblies **300** receive a corresponding clamping assembly (not shown) used to clamp the panel assemblies **300** together. The first alignment holes **376***a* of the panel assemblies **300** correspond to one of the alignment apertures **258** of the first end plate **218** and one of the alignment apertures **260** of the second end plate **220** (shown in FIGS. **2** and **4**). The second alignment holes **376***b* receive a corresponding clamping assembly (not shown). The second alignment holes **376***b* do not correspond to alignment apertures **258** and **260** in the end plates **218** and **220**, such that the clamping assemblies received by the second alignment holes **376***b* extend through the panel assemblies **300** but not the end plates **218** and **220**.

[0070] Still referring to FIGS. **10**A-**10**D, each of the corner flanges **364-370** includes flange collar **378** circumscribing the respective fluid port **340**, **342**, **346**, **348** defined in the corner flange. The flange collars **378** extend in the lateral direction from the first lateral face **305** of the frame **302**. Each corner flange **364-370** also includes a corresponding grooved mouth **380** that depends into the second lateral face **307** and circumscribes the respective fluid port **340**, **342**, **346**, **348** at the location laterally opposite the flange collar **378**. When the panel assemblies **300** are arranged in series and installed in the heat exchanger **200**, each flange collar **378** of the corner flanges **364-370** of the frame **302** of one of the panel assemblies **300** is received by one of the corresponding grooved mouths **380** of the corner flanges **364-370** of the frame **302** of a laterally adjacent panel assembly **300** to directly connect adjacent panel assemblies **300**. The flange collars **378** each include a set of piloting teeth **382** that facilitate aligning the flange collars **378** with the corresponding grooved mouths **380** and inserting the flange collars **378** therein. The piloting teeth **382** may include a piloting feature (e.g., a chamfer) for easier insertion of the teeth and the flange collar **378** into the corresponding grooved mouth **380**. Elastomeric seals (not shown), such as an O-ring, may be seated within each of the grooved mouths **380** and to create a fluid-tight seal between the adjacent panel assemblies **300** at the adjacent fluid ports **340**, **342**, **346**, **348** when the flange collars **378** are inserted into the grooved mouths **380**.

[0071] FIGS. **10**E and **10**F are enlarged views of the sections E and F, respectively, of the frame **302** in FIG. **8**, and depict two alignment features **388** of the frame that correspond to two alignment features **389** of each of the sheets or plates **326** and **328** (shown in FIG. **13**). The alignment features **388** of the frame **302** ("frame alignment features") are located proximate the first vertical end **304** and the second vertical end **306**, one of the frame alignment features **388** being located vertically above the header area **320** and the other frame alignment feature **388** being located vertically below the header area **322**. Each frame alignment feature **388** is also located at a longitudinal center of the frame **302**. As shown in FIG. **13**, the alignment features **389** of each sheet or plate **326**, **328** ("sheet alignment features") are at vertical and longitudinal locations of the plate that correspond to the vertical and longitudinal locations of the frame alignment features **388** when each plate **326**, **328** is positioned on the frame **302**. Pairs of corresponding frame and sheet alignment features **388**, **389** facilitate aligning each plate **326**, **328** to the frame **302** when attaching the plates to the frame. Each frame alignment feature **388** includes an alignment hole **390** that corresponds to an alignment hole **391** (FIG. **13**) of the corresponding sheet alignment feature **389**. The corresponding alignment holes **390**, **391** may receive alignment pins of a workstation (e.g., alignment pins **406** of the welding system **400** shown in FIG. **12**) for maintaining alignment and limiting movement between the frame **302** and the plate **326** or **328** when attaching the plate to the frame. The frame alignment features **388** and the sheet alignment features **389** may also be "flexible" and include respective flexures **392**, **393** that facilitate controlling alignment, tension, and tolerance stack between the frame **302** and the plate **326** or **328** when attaching the plate to the frame, as described in more detail below.

[0072] With additional reference to FIGS. **3** and **5**, and to FIG. **6** which shows a left elevation of the three-way heat exchanger **200** with various components omitted similar to

[0073] FIGS. **3** and **5**, the panel assemblies **300** are arranged in succession or series in the lateral direction as described above. In FIGS. **3**, **5**, and **6**, the plates **326** and **328** and the membranes **332** and **334** are omitted for convenience of illustration. When assembled and installed in the three-way heat exchanger **200**, for each pair of adjacent panel assemblies **300**, the membrane **332** of one of the panel assemblies **300** faces the membrane **334** of the other one of the panel assemblies **300**. The airflow gaps **216** are defined between the adjacent membranes **332** and **334**. Each panel assembly **300** may have a reduced width over the middle section **316**, such that the panel assemblies **300** are spaced apart over their adjacent middle sections **316** to define the airflow gaps **216**. Additionally and/or alternatively, the airflow gaps **216** may be defined and maintained by standoffs or spacers **386** between the middle sections **316** of adjacent panel assemblies **300**. The standoffs or spacers **386** extend in the lateral direction outward from the middle section **316** proximate the longitudinal ends **308** and **310**. The middle section **316** of each frame **302** may additionally and/or alternatively include the snap fittings **372** and the corresponding bores **374** (as shown in FIGS. **6** and **7**) proximate the longitudinal ends **308** and **310**. The snap fittings **372** and corresponding bores **374** located on the middle section **316** of the frame **302** may facilitate connecting adjacent panel assemblies **300** at the adjacent middle sections. The snap fittings **372** and corresponding bores **374** may be included in addition to the spacers **386**. Alternatively, the spacers **386** may in some examples be the snap fittings **372** which engage the bores **374** of the adjacent panel assembly **300** to connect the adjacent panel assemblies and maintain the width of the airflow gaps **216**.

[0074] The panel assemblies **300** are arranged in the three-way heat exchanger **200** such that, for each panel assembly, the first and second lateral faces **305** and **307** of the frame **302** are respectively oriented toward the first and second lateral sides **202** and **204** of the three-way heat exchanger **200**. The first and second longitudinal ends **308** and **310** are respectively located at the first and second longitudinal sides **206** and **208** of the three-way heat exchanger **200**, and the first and second vertical ends **304** and **306** are respectively located at the first and second vertical sides **210** and **212** of the three-way heat exchanger **200**.

[0075] The ports **340**, **342**, **346**, and **348** of the panel assemblies **300** align to define respective manifolds of the three-way heat exchanger **200** extending in the lateral direction through which heat transfer fluid and liquid desiccant flow to and from the panel assemblies **300** between the first and second lateral sides **202** and **204**. The liquid desiccant inlet ports **340** of the panel assemblies **300** align to form a liquid desiccant inlet manifold **242** that extends between the first and second lateral sides **202** and **204** proximate to the first vertical side **210** and the second longitudinal side **208** of the three-way heat exchanger **200**. The liquid desiccant outlet ports **342** of the panel assemblies **300** align to form a liquid desiccant outlet manifold **244** that extends between the first and second lateral sides **202** and **204** proximate to the second vertical side **212** and the first longitudinal side **206** of the three-way heat exchanger **200**. The heat transfer fluid inlet ports **346** of the panel assemblies **300** align to form a heat transfer fluid inlet manifold **246** that extends between the first and second lateral sides **202** and **204** proximate to the second vertical side **212** and the second longitudinal side **208** of the three-way heat exchanger **200**. The heat transfer fluid outlet ports **348** of the panel assemblies **300** align to form a heat transfer fluid outlet manifold **248** that extends between the first and second lateral sides **202** and **204** proximate to the first vertical side **210** and the first longitudinal side **206** of the three-way heat exchanger **200**.

[0076] The panel assemblies **300** may include O-rings or other elastomeric scaling members that form liquid-tight seals between the aligning ports **340**, **342**, **346**, and **348** of the adjacent panel assemblies to prevent fluid from leaking out of the respective manifolds **242-248**. For example, as described above, elastomeric seals (not shown), such as an O-ring, may be seated within each of the grooved mouths **380** and to create a fluid-tight seal between the adjacent panel assemblies **300** at the adjacent fluid ports **340**, **342**, **346**, **348** when the flange collars **378** are inserted into the grooved mouths **380**. In some examples, the elastomeric seals **384** are radial seals (e.g., radial O-ring seals). Moreover, in each of the corner flanges **364-370**, the snap fittings **372**, corresponding bores **374**, and the alignment holes **376** collectively surround the fluid port **340**, **342**, **346**, **348** defined in the corner flange, which may facilitate creating and maintaining a fluid-tight seal between the adjacent ports **340**, **342**, **346**, **348** that define the manifolds **242-248**.

[0077] As shown in FIGS. **3** and **5**, conduits **250**, **252**, **254**, and **256** are used to fluidly connect the heat transfer fluid inlet **232** and outlet **234** and the liquid desiccant inlet **236** and outlet **238** to a respective manifold for heat transfer fluid and liquid desiccant entering and exiting the three-way heat exchanger **200**. The liquid desiccant inlet **236** is fluidly connected to the liquid desiccant inlet manifold **242** by the conduit **250**. The liquid desiccant outlet **238** is fluidly connected to the liquid desiccant outlet manifold **244** by the conduit **252**. The heat transfer fluid inlet **232** is fluidly connected to the heat transfer fluid inlet manifold **246** by the conduit **254**. The heat transfer fluid outlet **234** is fluidly connected to the heat transfer fluid outlet manifold **248** by the conduit **256**. The conduits **250-256** may include any suitable fluid conduit (rigid and/or flexible) that enables heat transfer fluid and liquid desiccant to flow between the respective inlet and outlet and manifold, including, for example and without limitation, pipes, hoses, tubes, and combinations thereof. Each conduit **250-256** may be attached to the respective manifold **242-248** by coupling an end of the conduit to an end panel assembly **300** (i.e., the panel assembly **300** immediately adjacent to the lateral side **202** or **204**) at the appropriate one of the ports **340**, **342**, **346**, and **348**. The conduits **250-256** may be attached to the appropriate port **340**, **342**, **346**, and **348** of an end panel assembly **300** using any suitable means, include fasteners, threads, clamps, and the like.

[0078] The conduits **250** and **256** extend between the end plate **218** and the end panel assembly **300** at the first lateral side **202**. The conduits **252** and **254** extend between the end plate **220** and the end panel assembly **300** at the second lateral side **204**. The conduits **250** and **256** may extend through the end plate **218** to respectively define the inlet **236** or the outlet **234**, may be coupled to the respective inlet **236** or outlet **234** that is defined by the end plate **218**, or may be made integral with the end plate **218** and the respective inlet **236** or outlet **234** defined by the end plate **218**. The conduits **252** and **254** may extend through the end plate **220** to respectively define the outlet **238** or the inlet **232**, may be coupled to the respective outlet **238** or inlet **232** that is defined by the end plate **220**, or may be made integral with the end plate **220** and the respective outlet **238** or inlet **232** defined by the end plate **220**.

[0079] Each of the manifolds **242-248** may be closed at the lateral side **202** or **204** of the heat exchanger **200** opposite of the inlet or outlet to which the manifold is connected. The liquid desiccant inlet manifold **242** may be closed at the second lateral side **204**, the liquid desiccant outlet manifold **244** may be closed at the first lateral side **202** opposite the liquid desiccant inlet manifold **242**, the heat transfer fluid inlet manifold **246** may be closed at the first lateral side **202**, and the heat transfer fluid outlet manifold **248** may be closed at the second lateral side **204** opposite the heat transfer fluid inlet manifold **246**. The manifolds **242-248** may be closed at the respective lateral sides **202** or **204** by the end plate **218** or **220** (shown in FIGS. **2** and **4**). In particular, the end plate **218** may plug the ports **342** and **346** of the panel assembly **300** at the end of the panel assemblies **300** that is adjacent to the first lateral side **202** to close the manifolds **244** and **246** at the first lateral side **202**. The end plate **220** may plug the ports **340** and **348** of the panel assembly **300** at the end of the panel assemblies **300** that is adjacent to the second lateral side **204** to close the manifolds **244** and **246** at the second lateral side **204**. Additionally and/or alternatively, end caps or plugs **272** and **274** (see FIG. **11**) may be inserted into or otherwise disposed over the ports **342** and **346**, respectively, of the panel assembly **300** adjacent to the first lateral side **202** to close the manifolds **244** and **246** at the first lateral side **202**, and end caps or plugs **270** and **276** (see FIG. **11**) may be inserted into or otherwise disposed over the ports **340** and **348**, respectively, of the panel assembly **300** adjacent to the second lateral side **204** to close the manifolds **244** and **246** at the second lateral side **204**.

[0080] Referring now to FIG. **11**, operation of the three-way heat exchanger **200** will now be described. FIG. **11** is a schematic showing an interior view of the three-way heat exchanger **200** to depict flow of liquid desiccant and the heat transfer fluid through the manifolds **242-248** and the panel assemblies **300**. In the schematic of FIG. **11**, panel assemblies **300** are depicted with features exaggerated and/or simplified for convenience of illustration and description.

[0081] In an example operation of the heat exchanger **200**, an inlet air stream (e.g., the first or second inlet air stream **110** or **114** shown in FIG. **1**) enters via the airflow inlet **224** and flows through the air gaps **216** defined between adjacent panel assemblies **300** in the airflow direction **278**. The air flowing through the air gaps **216** is treated by liquid desiccant, indicated by the flow lines **344**, and heat transfer fluid, indicated by the flow lines **350**, that are channeled through each of the panel assemblies **300**. In some operations, the liquid desiccant **344** is concentrated liquid desiccant **152** from the liquid desiccant circuit **108** and the heat transfer fluid **350** is conditioner heat transfer fluid from the conditioner sub-system **104** shown in FIG. **1**, and the heat exchanger **200** is used to cool and dehumidify the air flowing through the air gaps **216**. In other operations, the liquid desiccant **344** is diluted liquid desiccant **154** from the liquid desiccant circuit **108** and the heat transfer fluid **350** is regenerator heat transfer fluid from the regenerator sub-system **106** shown in FIG. **1**, and the heat exchanger **200** is used to heat and reject moisture into the air flowing through the air gaps **216**.

[0082] The liquid desiccant **344** flows into the liquid desiccant inlet manifold **242** from the first lateral side **202**, via the liquid desiccant inlet **236** and the conduit **250** (shown in FIGS. **3-5**). The liquid desiccant **344** enters into the liquid desiccant header area **320** of each panel assembly **300** from the liquid desiccant inlet manifold **242** via the apertures **356**. In each panel assembly **300**, the liquid desiccant **344** flows from the liquid desiccant header area **320**, into the liquid desiccant channels **336** and **338** via the apertures **360** on each plate **326** and **328** (shown in FIGS. **7** and **8**), downward through the liquid desiccant channels **336** and **338**, and into the liquid desiccant header area **322** via the apertures **362** on each plate **326** and **328** (shown in FIGS. **7** and **8**). As the liquid desiccant **344** flows behind the membranes **332** and **334** of the panel assemblies **300**, the liquid desiccant **344** absorbs moisture from or desorbs moisture into the air flowing through the air gaps **216** adjacent to the membranes **332** and **334**. Moisture is permitted to permeate through each of the membranes **332** and **334** to enable the moisture to transfer between the liquid desiccant **344** and air in the air gaps **216**. The liquid desiccant **344**, having absorbed or desorbed moisture, exits each panel assembly **300** from the respective liquid desiccant header area **322** via the apertures **358**, and flows through the liquid desiccant outlet manifold **244** toward the second lateral side **204**. The liquid desiccant **344** exits the heat exchanger **200** via the conduit **252** and the liquid desiccant outlet **238** (shown in FIG. **5**).

[0083] The heat transfer fluid **350** flows into the heat transfer fluid inlet manifold **246** from the second lateral side **204**, via the heat transfer fluid inlet **232** and the conduit **254** (shown in FIGS. **4** and **5**). The heat transfer fluid **350** enters into each panel assembly **300** from the heat transfer fluid inlet manifold **246** via the apertures **352**. In each panel assembly **300**, the heat transfer fluid **350** flows upward through the heat transfer fluid channel **330**. The heat transfer fluid **350** flowing through the channel **330** is in thermal communication with the liquid desiccant **344** flowing through the liquid desiccant channels **336** and **338**. Heat is transferred between the heat transfer fluid **350** and the liquid desiccant **344** to remove heat from or reject heat into the air flowing through the air gaps **216**, depending on the operating mode of the heat exchanger **200**. The heat transfer fluid **350**, having absorbed or rejected heat, exits each panel assembly **300** via the apertures **354**, and flows through the heat transfer fluid outlet manifold **248** toward the first lateral side **202**. The heat transfer fluid **350** exits the heat exchanger **200** via the conduit **256** and the heat transfer fluid outlet **234** (shown in FIGS. **3-5**).

[0084] The direction of flow of the heat transfer fluid **350** and the liquid desiccant **344** in the illustrated embodiment is by way of example only, and may change in other embodiments of the heat exchanger **200**. For example, the liquid desiccant **344** may flow upward through the desiccant channels **336** and **338** on the panel assemblies **300**. In these examples, the direction of flow of the liquid desiccant **344** through the liquid desiccant inlet **236** and outlet **238** and the liquid desiccant inlet and outlet manifolds **242** and **244** would also be reversed. The heat transfer fluid **350** may flow downward through the heat transfer channels **330** of the panel assemblies **300**. In these examples, the direction of flow of the heat transfer fluid **350** through the heat transfer fluid inlet and outlets **232** and **234** and the heat transfer fluid inlet and outlet manifolds **246** and **248** would also be reversed. The liquid desiccant **344** and the heat transfer fluid **350** flow in counter-flow relation in the illustrated example, but may flow in the same direction through the panel assemblies in alternative examples.

[0085] Referring now to FIGS. **12-16**, example systems and methods for assembling one or more of the panel assemblies **300** (also referred to as a multilayer panel) will now be described. FIG. **12** is a schematic of an example system **400** that may be used to attach each plate **326**, **328** (also referred to as heat exchange sheets **326**, **328** or sheets **326**, **328**) to the frame **302** and/or each membrane **332**, **334** to the respective sheet **326**, **328**. The welding system **400** includes a work chamber **402** and a work platform **404** positioned in the chamber **402**. The work platform **404** is sized and shaped to support the frame **302**, the plates **326**, **328**, and the membranes **332**, **334**. The work platform **404** includes two alignment pins **406** that are sized, shaped, and positioned on the work platform to be inserted into the two pairs of corresponding frame alignment features **388** (FIGS. **10**E and **10**F) and sheet alignment features **389** (FIG. **13**). FIG. **12** schematically depicts the frame **302** positioned on the work platform **404** and the heat exchange sheet **326** or **328** positioned on the frame, with the alignment pins **406** extending through the corresponding frame and sheet alignment features **388**, **389**. In particular, each alignment pin **406** extends through a corresponding pair of an alignment hole **390** of the frame **302** and an alignment hole **391** of the sheet **326** or **328**. Inserting the alignment pins **406** through the pairs of corresponding pairs alignment holes **390**, **391** operates to maintain alignment and limit relative movement between the frame **302** and the sheet **326** or **328** when the system **400** is used to attach (e.g., weld) the sheet **326** or **328** to the frame **302**.

[0086] In the example illustrated in FIG. **12**, one sheet **326** or **328** having the membrane **332** or **334** previously attached thereto is positioned on a lateral face **305** and **307** of the frame **302** for attaching the sheet **326** or **328** to the frame **302**. This is for example only, according to one example method of assembling the panel assembly **300** using the system **400**. In other methods, the frame **302** may have the sheet **326** or **328** positioned thereon without the membrane **332** or **334** having been previously attached to the sheet. The system **400** may additionally and/or alternatively be used to attach (e.g., heat seal) the membrane **332** or **334** to the sheet **326** or **328** before attaching the sheet **326** or **328** to the frame **302**. In such methods, the sheet **326** or **328** may be positioned on the work platform **404** with the alignment pins **406** extending through the alignment holes **391** and the membrane **332** or **334** is positioned across the sheet **326** or **328** between the alignment holes **391**, without inserting the alignment pins **406** through the membrane **332** or **334**. In some examples, the membrane **332** or **334** may also include alignment features corresponding to the sheet alignment features **388** for inserting the alignment pins therethrough. Both sheets **326** and **328**, with or without the respective membranes **332** and **334** previously attached, may be attached (e.g., welded) to the frame **302** using the system **400**. In such methods, the first sheet **326** or **328** is attached to the frame **302** with the first sheet **326** or **328** positioned on the frame **302** as shown in FIG. **12**, and the second sheet **328** or **326** is attached to the frame **302** with the frame **302** flipped (i.e., rotated 180° about a vertical axis extending between the vertical ends **304** and **306**) such that the first sheet **326** or **328** previously attached (and, optionally, the membrane **332** or **334**) is sandwiched between the frame and the work platform **404** and the second sheet **328** or **326** is positioned on the other lateral face **307** or **305** of the frame **302**. The alignment pins **406** may extend through the alignment holes **388** of the frame **302** and the alignment holes **389** of each sheet **326** and **328** respectively positioned on the lateral faces **305** and **307** of the frame **302**.

[0087] Referring to FIGS. **10**E-**10**F and **13**, the frame alignment features **388** and the sheet alignment features **389** are flexible and include respective flexures **392**, **393** that facilitate controlling alignment, tension, and tolerance stack between the frame **302** and the sheet **326** or **328** when attaching the plate to the frame. The flexures **392**, **393** allow the respective alignment holes **390**, **391** to move or flex. The flexures **392** of the frame **302** (FIGS. **10**E and **10**F) are each formed by two cut-outs **394** that are located above and below the respective alignment hole **390** and define a “bow tie” shape. The flexure **392** formed by the two cut-outs **394** is a filament in which the alignment hole **390** is defined and allows the alignment hole to move or flex within the two cut-outs **394**. The flexures **393** of the sheets **326** and **328** (FIG. **13**) are each formed by two slits **395** located on each side of the alignment hole **391**. The flexure **393** formed by the two slits **395** is a flexible region radiating out from the alignment hole **391** that allows the alignment hole **391** to move or flex.

[0088] Referring to FIG. **12**, the system **400** also includes an attachment tool **408** (e.g., a laser welder or a heat sealer) operably positioned in the chamber **402**. The tool **408** includes a working element **410** (e.g., a heater or a laser) connected to a support **412**. The support **412** is connected to an arm **414** (or multiple arms **414**) of the tool **408** that operates to move the support **412** and the working element **410** within the chamber **402** relative to the work platform **404**. Movement of the arm **414** may be facilitated by any suitable means, such as linear actuators, motors, hydraulic cylinders, pneumatic cylinders, and the like. Movement of the arm **414** controls movement of the support **412** and the working element **410** for attaching the sheets **326** and **328** to the frame **302** and/or attaching the membranes **332** and **334** to the respective sheets **326** and **328**. In the example system **400**, the arm **414** is operable to move the support **412** and the working clement **410** downward toward and upward away from the work platform **404**. The arm **414** may additionally and/or alternatively move the support **412** and working element **410** in another direction. The arm **414** may be operable for multi-axis movement in some examples. The working platform **404** may additionally and/or alternatively be operably coupled to one or more actuators (e.g., linear actuators, motors, hydraulic cylinders, pneumatic cylinders, and the like) for controlling movement, along a single axis or multiple axes, of the working platform **404** relative to the support **412** and the working element **410**.

[0089] The working element **410** (e.g., heater or laser) is operable to form suitable seams or patterns for attaching (e.g., welding) the sheets **326** and **328** to the frame **302** and/or to for attaching (e.g., heat sealing) the membranes **332** and **334** to the respective sheets **326** and **328**. An example heat seal pattern between the membrane **332** or **334** and the sheet **326** or **328** is shown in FIG. **14** and an example welding pattern between the sheet **326** or **328** and the frame **302** is shown in FIG. **15**. Attaching the sheets **326** and **328** to the frame **302** may be performed using the same tool **408** as is used to attach the membranes **332** and **334** to the respective sheets **326** and **328**, or different tools **408** may be used for these operations. Where different tools **408** are used, the different tools **408** may be operated in the same or a different chamber **402**. Suitably, the working element **410** of each tool **408** is operable to form the welds and/or the heat seals in a single operation. Alternatively stated, the working element **410** of the tool **408** (e.g., a laser welder or a heat sealer) may “stamp” or “punch” the welds or heat seals in one simultaneous stroke, rather than tracing out the weld patterns or forming the heat seals individually. Forming the welds or heat seals in a single operation may significantly reduce time and costs associated with attaching the membranes **332**, **334** to the sheets **326**, **328** and/or attaching the sheets **326**, **328** to the frame **302**. The alignment holes **390** and **391** and the flexures **392** and **393** may facilitate precise alignment and suitable tension of the frame **302** and the sheets **326** and **328** to enable the welds or heat seals to be formed in the single operation as described.

[0090] The system **400** also includes a controller **416** connected in communication with the attachment tool **408**, and configured to control operation of the tool **408** (e.g., by controlling operation of the arm **414** and/or the working element **410**). For example, the controller **416** may be configured to control operation of the arm **414** to move the support **412** and the working element **410** towards and away from the work platform **404**. The controller **416** may additionally and/or alternatively be configured to control operation of the working element **410** for forming a suitable weld or heat seal pattern, depending on whether the tool **408** is being used to perform a welding operation to attach the sheets **326** and **328** to the frame **302** and/or to attach the membranes **332** and **334** to the respective sheets **326** and **328**.

[0091] The controller **416** may include any suitable computer and/or other processing unit, including any suitable combination of computers, processing units and/or the like that may be communicatively connected to one another and that may be operated independently or in connection within one another (e.g., the controller **416** may form all or part of a controller network). The controller **416** may include one or more modules or devices, one or more of which is enclosed within a single housing, or may be located remote from one another. The controller **416** may include one or more processor(s) and associated memory device(s) configured to perform a variety of computer-implemented functions (e.g., performing the functions disclosed herein). As used herein, the term “processor” refers not only to integrated circuits, but also refers to a controller, a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific integrated circuit, and other programmable circuits. Additionally, memory device(s) of the controller **416** may be or include memory element(s) including, but not limited to, computer readable medium (e.g., random access memory (RAM)), computer readable non-volatile medium (e.g., a flash memory), a floppy disk, a compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), a digital versatile disc (DVD) and/or other suitable memory elements. Such memory device(s) may be configured to store suitable computer-readable instructions that, when implemented by the processor(s), configure or cause the controller **416** to perform various functions described herein including, but not limited to, controlling the tool **408**.

[0092] The controller **416** may communicate with one or more components of the tool **408** via a communication interface coupled in communication with one or more of these devices. The communication interface may include, without limitation, a wired network adapter, a wireless network adapter, a mobile telecommunications adapter, a serial communication adapter, or a parallel communication adapter. The communication interface may receive a data signal from, or transmit a data signal to, one or more remote devices, such as the tool **408**. The controller **416** may also include a presentation interface coupled to one or more of the processors. The presentation interface may present information, such as a user interface, to an operator of the system **400**. In one embodiment, the presentation interface includes a display adapter (not shown) that is coupled to a display device (not shown), such as a cathode ray tube (CRT), a liquid crystal display (LCD), an organic LED (OLED) display, or an “electronic ink” display. In some embodiments, the presentation interface includes one or more display devices. In addition, or alternatively, the presentation interface includes an audio output device (not shown), for example, without limitation, an audio adapter, a speaker, or a printer (not shown). The controller **416** may also include a user input interface coupled to one or more of the processors and operable to receive input from the operator. The user input interface may include, for example, and without limitation, a keyboard, a pointing device, a mouse, a stylus, one or more input buttons, a touch sensitive panel, such as, without limitation, a touch pad or a touch screen, and/or an audio input interface, such as, without limitation, a microphone. A single component, such as a touch screen, may function as both a display device of the presentation interface and the user input interface.

[0093] FIGS. **13-15** depict a sequence of operations that may be performed to assemble the multilayer panel **300** (shown in FIG. **7**) in accordance with a method **500** shown in FIG. **16**. The method **500** may be performed using one or more systems that include a work platform and a suitable attachment tool (e.g., one or more of the system **400** shown in FIG. **12**). FIG. **13** is a schematic of a heat exchange sheet **326**, **328** included in the panel assembly **300**. FIG. **14** is a schematic of a membrane **332**, **334** attached to the sheet **326** or **328** of FIG. **13**. FIG. **15** is a schematic of the sheet **326**, **328**, with the membrane **332**, **334** of FIG. **14** attached to the frame **302**. The sequence of operations shown in FIG. **13-15** and the method **500** of FIG. **16** are provided by way of example only, and the operations described may be performed in any suitable order. For example, the sheet **326** or **328** may be attached to the frame **302** before the membrane **332** or **334** is attached to the sheet **326** or **328**. The operations may also be repeated for both heat exchange sheets **326** and **328** and membranes **332** and **334** to assemble the multilayer panel **300** including two sheets **326** and **328**, one sheet **326**, **328** attached to each lateral face **305**, **307** of the frame **302**, and two membranes **332**, **334**, one membrane **332**, **334** attached to each sheet **326**, **328**.

[0094] The method **500** includes providing **502** the heat exchange sheet **326**, **328** shown in FIG. **13** that includes the two alignment features **389** and the two series of apertures **360**, **362**. The two series of apertures **360**, **362** are located between the two alignment features **389**. One alignment feature **389** is located adjacent each vertical end of the sheet **326**, **328** (relative to the orientation shown in FIG. **13**). The series of apertures **360** is located proximate the “top” alignment feature **389** and the series of apertures **362** is located proximate the “bottom” alignment feature **389**.

[0095] The method **500** also includes positioning **504** the heat exchange sheet **326**, **328** on a work platform (e.g., the work platform **404** of FIG. **12**) such that two alignment pins (e.g., the alignment pins **406**) are inserted through the two alignment holes **391** of the heat exchange sheet **326**, **328**. The method **500** also includes attaching (e.g., heat sealing) **506** the membrane **332**, **334** to the heat exchange sheet **326**, **328** positioned on the work platform, such that a desiccant channel **336**, **338** (FIG. **9**) is defined between the membrane **332**, **334** and the sheet **326**, **328**. As shown in FIG. **14**, the membrane **332**, **334** is shorter in height than the sheet **326**, **328**, and is positioned on the sheet such that the membrane overlaps the two series of apertures **360**, **362** and the alignment features **389** are uncovered by the membrane. The membrane **332**, **334** does not include alignment features that correspond to the alignment features **389** of the sheet **326**, **328** in this example. The membrane **332**, **334** may be positioned and aligned on the sheet by visual inspection such that the apertures **360**, **362** are sufficiently overlapped by the membrane, and the alignment features **389** remain uncovered. In other examples, the membrane **332**, **334** may include alignment features that facilitate centering the membrane on the sheet **326**, **328**.

[0096] Attaching **506** the membrane **332**, **334** to the heat exchange sheet **326**, **328** may be performed using a heat sealing tool (e.g., the tool **408**) that operates to heat seal the membrane to the sheet along various “heat seals” (e.g., lines, seams, patterns) as shown in FIG. **14**. In the illustrated example, two heat seal lines **418**, **420** are respectively formed adjacent the two series of apertures **360**, **362**, such that the desiccant channel **336**, **338** is connected to both series of apertures **360**, **362** and is sealed vertically above the apertures **360** and vertically below the apertures **362**. Each heat seal line **418**, **420** includes arcs that complement the apertures **360**, **362**, respectively, and horizontal (longitudinal) segments connecting each arc.

[0097] Each aperture **360**, **362** is centered within a respective arc of the heat seal lines **418**, **420**, which provides a space for desiccant to flow through each aperture **360**, **362** between the sheet **326**, **328** and the membrane **332**, **334**. Misalignment between the apertures **360**, **362** and the heat seal lines **418**, **420** may result in the apertures **360** and/or **362** being too close to, or partially covered, by the heat seal lines **418** and/or **420**, which is disadvantageous because it restricts or impedes desiccant flow through the desiccant channel **336**, **338**, increases tensile stress in the membrane **332**, **334**, and/or otherwise negatively impacts performance of the panel assembly **300**. Thus, in one example of heat sealing, alignment between the heat seal lines **418**, **420** and the series of apertures **360**, **362**, and more particularly the centering of the apertures within the arcs of the heat seal lines, is controlled using the alignment features **389** of the heat exchange sheet **326**, **328**. That is, the alignment pins **406** that are inserted in the alignment holes **391** of the sheet **326**, **328** operate to control alignment between the sheet and the heat sealing tool to ensure that the arcs of the heat seal lines **418**, **420** and the respective apertures **360**, **362** remain centered.

[0098] The flexures **393** may also operate to control alignment between the sheet **326**, **328** and the heat sealing tool, as well as tension within the sheet, when positioned on the work platform. In particular, the flexures **393** compensate for dimensional variations (e.g., height variations) in the sheet **326**, **328** that may result, for example, from manufacturing tolerances and cause variations in the vertical spacing between the alignment holes **391**. The likelihood for such variations may be exacerbated or changed depending, for example, on the material used for the sheet **326**, **328**. For example, thermoplastic or polymer materials (e.g., polyolefins such as polypropylene and/or polyethylene) that may be used to make the sheet **326**, **328** may exhibit a large variation in dimensions. Such variations may result in undesired tension or stress on the sheet **326**, **328** when the sheet **326**, **328** is positioned on the work platform **404** and the alignment pins **406** are inserted in the alignment holes **391**, which may lead to deformation of the sheet **326**, **328** (e.g., bowing or stretching) and misalignment between the sheet **326**, **328** and the heat sealing tool. The flexures **393** enable the alignment holes **391** to flex, which facilitates reducing the negative impact that manufacturing tolerances may have on the alignment between the sheet **326**, **328** and the heat sealing tool by reducing or eliminating any deformation in the sheet in the areas that require precise alignment with the heat sealing tool (e.g., proximate the series of apertures **360**, **362**).

[0099] FIG. **14** shows additional heat seals that may be formed between the sheet **326**, **328** and the membrane **332**, **334**. For example, heat seal “dashes” **422**, **424** may be formed at opposite vertical ends of the membrane **332**, **334**. The heat seal dashes **422** are formed above the heat seal line **418** and the heat seal dashes **424** are formed below the heat seal line **420**. The heat seal dashes **422**, **424** may facilitate retaining “excess” portions of the membrane **332**, **334** above and below the heat seal line **418**, **420**, respectively. The heat seal dashes **422**, **424** may also facilitate increasing the reliability of the attachment between the membrane **332**, **334** and the sheet **326**, **328**. The dashes **422**, **424** may be formed at longitudinal locations corresponding to the longitudinal location of the arcs of the heat seal lines **418**, **420**. Discrete heat seals or heat seal “dots” **426** may also be formed between the heat seal lines **418** and **420**. The heat seal dots **426** may define discrete flow paths for the desiccant flowing between the sheet **326**, **328** and the membrane **332**, **334**. Any suitable number and/or arrangement of the heat seal dots **426** may be implemented. The number and/or location of the heat seal dots **426** may be selected to facilitate controlling distribution and/or a flow rate of the desiccant through the desiccant channel **336**, **338**. Additional detail on suitable patterns for the heat seal dots **426** is described in U.S. Pat. No. 10,921,001, issued on Feb. 16, 2021, which is incorporated by reference in its entirety.

[0100] As described above, the heat sealing tool (e.g., the tool **408**) may be operable to heat seal the membrane **332**, **334** to the heat exchange sheet **326**, **328** in a single heat seal operation, such that the heat sealing tool stamps or punches each of the heat seals (e.g., the heat seal lines **418**, **420**, the heat seal dashes **422**, **424**, and the heal seal dots **426**) in one stroke. This requires precise alignment between the heat sealing tool and the sheet **326**, **328** to ensure that the heat seals are not misaligned with the apertures **360**, **362**. Suitably, the flexible alignment features **389** of the sheet **326**, **328**, with the alignment pins **406** of the work platform **404** inserted therethrough, operate to control alignment, tension, tolerance, and movement of the sheet relative to the platform and the heat sealing tool such that the heat sealing can be performed in a single operation.

[0101] Referring to FIG. **16**, the method **500** also includes positioning **508** the frame **302** on a work platform (e.g., the work platform **404** of FIG. **12**), which may be the same work platform on which the heat exchange sheet **326**, **328** is positioned **504** before attaching the membrane **332**, **334** to the sheet or may be a different work platform. The frame **302** is positioned **508** on the work platform **404** such that two alignment pins (e.g., the alignment pins **406**) are inserted through the two alignment holes **390** (FIGS. **10**E and **10**F) of the frame **302**. The method **500** also includes positioning **510** the heat exchange sheet **326**, **328** on the frame **302**. The sheet **326**, **328** is positioned **510** on the frame **302** such that the sheet extends across the middle section **316** and the header sections **312** and **314** of the frame, and such that the sheet alignment features **389** are aligned with the corresponding frame alignment features **388** (shown in FIG. **15**). The alignment pins **406** inserted through the two alignment holes **390** of the frame **302** are also inserted through the corresponding alignment holes **391** of the sheet **326**, **328** positioned **510** on the frame **302** to facilitate controlling alignment between the sheet and the frame and limit relative movement therebetween.

[0102] The method **500** also includes attaching (e.g., laser welding) **512** the sheet **326**, **328** to the frame **302**. The heat exchange sheet **326**, **328** may be positioned **510** on and attached **512** to the frame **302** after attaching **506** the membrane **332**, **334** to the sheet, or before the membrane is attached **506**. Attaching **512** the sheet **326**, **328** to the frame **302** may be performed using a laser welding tool (e.g., the tool **408**) that operates to laser weld the sheet to the frame along various "welds" (e.g., paths or seams) as shown in FIG. **15**. In the illustrated example, the sheet **326**, **328** is attached to the middle section **316** of the frame **302** along a middle weld **428**, the first header section **312** of the frame along a first outer weld **430** (or "top" weld), and the second header section **314** of the frame along a second outer weld **432** (or "bottom" weld). The middle weld **428** envelops and seals the sheet **326**, **328** around the heat transfer fluid area **324** defined by the frame **302**. The top weld **430** envelops and seals the sheet **326**, **328** around the first liquid desiccant header area **320** defined by the frame **302**. The bottom weld **432** envelops and seals the sheet **326**, **328** around the second liquid desiccant header area **322** defined by the frame **302**. The heat transfer fluid area **324** and the liquid desiccant header areas **320**, **322** are shown, for example, in FIG. **8**. The series of apertures **360** of the sheet **326**, **328** are enveloped by the top weld **430** and aligned with the liquid desiccant header area **320**, and the series of apertures **362** of the sheet are enveloped by the bottom weld **432** and aligned with the liquid desiccant header area **322**. The apertures **360**, **362** connect the desiccant channel **336**, **338** defined between the membrane **332**, **334** and the sheet **326**, **328** with the liquid desiccant header areas **320**, **322**, and the region of the sheet located between the apertures **360**, **362** and sealed to the middle section **316** by the middle weld **428** separates the desiccant channel from the heat transfer fluid area **324**. The welds **428**, **430**, and **432** also operate to restrict flow of heat transfer fluid into the liquid desiccant header areas **320**, **322** and restrict flow of liquid desiccant into the heat transfer fluid area **324** or channel **330** (shown in FIG. **9**).

[0103] As shown in FIG. **15**, the heat seals (e.g., the heat seal lines **418**, **420**, the heat seal dashes **422**, **424**, and the heat seal dots **426**) used to attach **506** the membrane **332**, **334** to the sheet **326**, **328** have minimal overlap with the welds **428**, **430**, **432** used to attach **512** the sheet to the frame **302**. The heat seal line **418** and heat seal dashes **422** are substantially surrounded by the top weld **430**, the heat seal dots **426** are substantially surrounded by the middle weld **428**, and the heat seal line **420** and heat seal dashes **424** are substantially surrounded by the bottom weld **428.** This may suitably minimize energy applied to the membrane **332**, **334** and the sheet **326**, **328** when assembling the multilayer panel **300**. The welds **428**, **430**, and **432** may also be substantially formed between the sheet **326**, **328** and the frame **302** only, to reduce or eliminate weld seams formed with the membrane **332**, **334**. FIG. **15** depicts internal weld portions of the welds **428**, **430**, **432** (i.e., portions of the welds covered by the membrane **332**, **334**) in dashed lines. In the illustrated example, the internal weld portions of the welds **428**, **430**, **432** are on the frame bars **325**, **327** (shown in FIG. **8**) of the frame **302**, which separate the heat transfer fluid area **324** from the liquid desiccant header areas **320**, **322**, respectively. The internal weld portions of the welds **428**, **430**, **432** may be formed between the sheet **326**, **328** and the frame **302** only. This may prevent creating welds that extend across the desiccant channel **336**, **338** below the apertures **360** and/or above the apertures **362** and between the membrane **332**, **334** and the sheet **326**, **328**, which may otherwise restrict or impede flow of the liquid desiccant in the desiccant channel **336**, **338** between the header areas **320** and **322**.

[0104] As described above, the membrane **332**, **334** may be made from a translucent material, such as a translucent thermoplastic or polymer (e.g., polypropylene), that has a suitable laser transmissivity to enable light emitted from a laser (e.g., the working clement **410**) to pass therethrough. The plate **326**, **328** may also be made from a translucent thermoplastic or polymer with laser transmissivity. Laser transmissivity of the membrane **332**, **334** and/or frame **326**, **328** may additionally and/or alternatively be facilitated using translucent coating (e.g., oil) that may be subsequently removed after welding. The frame **302** may include laser-absorbing additives (e.g., carbon black) that enable the frame to absorb light emitted from the laser, heating and melting the frame at suitable locations such that, when the frame cools and solidifies, the welds **428**, **430**, and **432** are formed. Additionally, and/or alternatively, the frame **302** may be made from a laser-absorbing thermoplastic or polymer. The translucent membrane **332**, **334** and sheet **326**, **328** enable laser welding the frame **302** and the sheet with the membrane **332**, **334** attached (e.g., heat sealed) to the plate, without forming a weld between the membrane and the sheet. The laser (e.g., the working element **410**) may be operated to emit light at a suitable wavelength selected based on the transmissivity of the translucent membrane **332**, **334** and sheet **326**, **328** and the laser-absorption capability of the frame **302**.

[0105] Inserting the alignment pins **406** through the pairs of corresponding frame and sheet alignment features **388**, **389** when the sheet **326**, **328** is positioned **510** on and attached **512** to the frame **302** suitably facilitates controlling alignment of the sheet and the frame and preventing "mis-welds." In particular, the pairs of corresponding frame and sheet alignment features **388**, **389**, with the alignment pins **406** inserted therein, ensure that the welds **428**, **430**, and **432** are properly formed to seal the sheet **326**, **328** around the heat transfer fluid area **324** and the liquid desiccant header areas **320**, **322** thereby preventing against flow of heat transfer fluid entering the desiccant channel **336**, **338** and/or flow of liquid desiccant entering the heat transfer fluid channel **330**. Controlling alignment between the sheet **326**, **328** and the frame **302** also suitably facilitates ensuring that the apertures **360**, **362** are properly aligned with the liquid desiccant header areas **320**, **322**, respectively, and that there is minimal overlap between the welds used to attach **512** the sheet to the frame and the heat seals used to attach **506** the membrane **332**, **334** to the sheet.

[0106] The frame and sheet flexures **392**, **393** may also operate to control alignment between the sheet **326**, **328**, the frame **302**, and the laser welding tool as well as tension in the sheet and the frame when positioned on the work platform. As described above, the sheet flexures **393** (FIG. **13**) compensate for dimensional variations (e.g., height variations) in the sheet **326**, **328** that may result from manufacturing tolerances and cause variations in the vertical spacing between the alignment holes **391**. The frame flexures **392** (FIGS. **10**E and **10**F) also operate to compensate for dimensional variations (e.g., height variations) in the frame **302** that may result from manufacturing tolerances and cause variations in the vertical spacing between the alignment holes **390**. As described above, the propensity for such variations may be exacerbated, for example, when thermoplastic or polymer materials (e.g., polyolefins such as polypropylene and/or polyethylene) are used to make the sheet **326**, **328** and/or the frame **302**. The sheet and frame flexures **392**, **393** enable the respective alignment holes **390**, **391** to flex, which facilitates reducing the negative impact that manufacturing tolerances may have on the alignment between the sheet **326**, **328**, the frame **302**, and the laser welding tool by reducing or eliminating any deformation (e.g., bending or stretching) in the sheet and frame in the areas that require precise alignment with the laser welding tool. Furthermore, the frame flexures **392** and the sheet flexures **393** independently operate (i.e., allow the respective alignment hole **390**, **391** to flex independent of the other alignment hole **391**, **390**) to facilitate controlling tolerance stack between variations in the dimensions of the frame and the sheet.

[0107] The laser welding tool (e.g., the tool **408**) may be operable to weld the sheet **326**, **328** to the frame **302** in a single welding operation, such that the laser welding tool forms, stamps or punches each of the welds **428**, **430**, **432** in one simultaneous stroke. This requires precise alignment between the laser welding tool, the sheet **326**, **328**, and the frame **302** to ensure that the welds are not misaligned. Suitably, the flexible frame and sheet alignment features **388**, **389**, with the alignment pins **406** of the work platform **404** inserted through corresponding pairs thereof, operate to control alignment, tension, tolerance stack, and movement between the frame **302**, the sheet **326**,

328, and the laser welding tool such that the laser welding can be performed in a single operation.
[0108] Example systems and methods described include assembling multilayer panels suitable for use in three-way heat exchangers for removing heat and moisture from a flow of air and/or rejecting heat and moisture into a flow of air. An example multilayer panel includes a frame, two sheets attached to the frame to define a heat transfer fluid channel, and a membrane attached to each sheet to define desiccant channels. The manufacturability and assembly of the multilayer panel is improved by flexible alignment features of the frame and the sheets that facilitate controlling alignment and manufacturing tolerances when attaching the membranes to the respective sheets and when attaching the sheets to the frame. The flexible alignment features enable precise alignment which facilitates attaching the membranes to the sheets using a single operation and attaching the sheets to the frame using a single operation. Heat seals are advantageously formed between the membranes and the respective sheets for facilitating controlling flow and/or distribution of the liquid desiccant through the desiccant channels. Welds are advantageously formed between the sheets and the frames to facilitate preventing against flow of heat transfer fluid and liquid desiccant in undesired areas of the multilayer panel. The heat seals and welds are also suitably located to minimize overlap between the heat seals and the welds and minimize energy applied to the sheets and the membranes.
[0109] Embodiments of HVAC systems and methods of operating the systems are described above in detail. The systems and methods are not limited to the specific embodiments described herein, but rather, components of the system and methods may be used independently and separately from other components described herein. For example, the systems and methods described herein may be used in systems other than HVAC systems.
[0110] When introducing elements of the present disclosure or the embodiment(s) thereof, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” “containing” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The use of terms indicating a particular orientation (e.g., “top”, “bottom”, “side”, “vertical”, “lateral”, “longitudinal”, etc.) is for convenience of description and does not require any particular orientation of the item described.
[0111] The terms “about,” “substantially,” “essentially” and “approximately,” and their equivalents, when used in conjunction with ranges of dimensions, concentrations, temperatures or other physical or chemical properties or characteristics is meant to cover variations that may exist in the upper and/or lower limits of the ranges of the properties or characteristics, including, for example, variations resulting from rounding, measurement methodology or other statistical variation.
[0112] As various changes could be made in the above constructions and methods without departing from the scope of the disclosure, all matter contained in the above description and shown in the accompanying drawing(s) shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A method of assembling a multilayer panel for use in a heat exchanger, the method comprising: positioning a frame on a work platform, the frame having two header sections and a middle section, the middle section defining a heat transfer fluid area and each header section defining a header area; positioning a heat exchange sheet on the frame across the middle section and each header section; welding the heat exchange sheet to the frame to form a middle weld enveloping the heat transfer fluid area and two outer welds each enveloping a corresponding one of the header areas; and controlling alignment between the frame and the heat exchange sheet during welding.
2. The method of claim 1, wherein controlling alignment between the frame and the heat exchange sheet during welding includes inserting alignment pins of the work platform through corresponding frame alignment features and heat exchange sheet alignment features when positioning the heat

exchange sheet on the frame.
3. The method of claim 2, further comprising controlling tolerance stack between the frame and the heat exchange sheet using flexures at each of the corresponding frame and the heat exchange sheet alignment features.
4. The method of claim 1, wherein the heat exchange sheet has two series of apertures, wherein each of the two series of apertures are aligned with one of the headers of the frame, wherein the method further comprises attaching a membrane to the heat exchange sheet to define a desiccant channel connected to each of the two header areas by the two series of apertures.
5. The method of claim 4, wherein attaching the membrane to the heat exchange sheet comprises heat sealing the membrane to the heat exchange sheet to form heat seal lines adjacent each series of apertures, and wherein the method further comprises controlling alignment between the heat seal lines and the series of apertures during heat sealing using flexible alignment features of the heat exchange sheet.
6. The method of claim 1, wherein positioning the frame on the work platform comprises inserting an alignment pin of the work platform into a frame alignment feature, the frame alignment feature including a flexure defining a flexible region on the frame.
7. The method of claim 6, wherein positioning the frame on the work platform comprises inserting the alignment pin of the work platform into an alignment hole of the flexure formed within the frame, wherein the flexure is formed by at least two cut-outs straddling the alignment hole.
8. The method of claim 1, wherein positioning the heat exchange sheet on the frame comprises inserting an alignment pin of the work platform into a heat exchange sheet alignment feature, the heat exchange alignment feature including a flexure defining a flexible region on the heat exchange sheet.
9. The method of claim 8, wherein positioning the heat exchange sheet on the frame comprises inserting the alignment pin of the work platform into an alignment hole of the flexure formed within the heat exchange sheet, wherein the flexure is formed by at least two slits formed on opposing sides of the alignment hole.
10. The method of claim 8, wherein positioning the heat exchange sheet on the frame comprises inserting the alignment pin of the work platform into a frame alignment feature and the heat exchange sheet alignment feature, and controlling alignment between the frame and the heat exchange sheet during welding comprises controlling alignment between the frame and the heat exchange sheet during welding using the flexure formed within the heat exchange sheet and the frame alignment feature.
11. The method of claim 1, wherein positioning the heat exchange sheet on the frame comprises aligning each of two series of apertures of the heat exchange sheet with one of the header areas, wherein each of the two series of apertures are located proximate one of two heat exchange sheet alignment features.
12. The method of claim 11, further comprising attaching a membrane to the heat exchange sheet.
13. The method of claim 12, wherein attaching the membrane to the heat exchange sheet is performed prior to welding the heat exchange sheet to the frame.
14. The method of claim 13, wherein attaching the membrane to the heat exchange sheet comprises forming heat seal lines adjacent each of the series of apertures to define a desiccant channel extending between and connected to the two series of apertures, wherein alignment between the heat seal lines and the series of apertures is controlled by alignment pins received within the corresponding heat exchange sheet alignment features.
15. The method of claim 14, wherein alignment between the heat seal lines and the series of apertures is controlled by the alignment pins received within a flexure of the heat exchange sheet alignment feature formed within the heat exchange sheet.
16. A method of assembling a multilayer panel for use in a heat exchanger, the method comprising: positioning a frame on a work platform, the frame having two header sections and a middle section,

the middle section defining a heat transfer fluid area and each header section defining a header area; positioning a heat exchange sheet on the frame across the middle section and each header section, the heat exchange sheet having a sheet alignment feature including a flexure formed within the heat exchange sheet, the flexure defining a flexible region on the heat exchange sheet; welding the heat exchange sheet to the frame to form a middle weld enveloping the heat transfer fluid area and two outer welds each enveloping a corresponding one of the header areas; and controlling alignment between the frame and the heat exchanger sheet during welding using the flexure formed within the heat exchange sheet.

17. The method of claim 16, wherein controlling alignment between the frame and the heat exchange sheet during welding comprises controlling alignment between the frame and the heat exchange sheet during welding using an alignment pin received within the flexure formed within the heat exchange sheet.

18. The method of claim 17, wherein positioning the heat exchange sheet on the frame comprises aligning each of two series of apertures of the heat exchange sheet with one of the header areas, wherein each of the two series of apertures are located proximate one of two exchange sheet alignment features.

19. The method of claim 16, wherein positioning the frame on the work platform comprises each header section defining a frame alignment feature, wherein controlling alignment between the frame and the heat exchange sheet during welding comprises using the flexure formed within the heat exchange sheet and at least one of the frame alignment features.

20. The method of claim 19, wherein positioning the frame on the work platform comprises inserting the alignment pin of the work platform into an alignment hole of a flexure at a corresponding frame alignment feature, wherein the flexure is formed by at least two cut-outs straddling the alignment hole.